Stratocracy - Prototype GDD

Stratocracy — Prototype GDD

Hex turn-based strategy · Unreal Engine 5.8 · 7-week AI-agent jam

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1. Executive Summary

Concept. Two commanders battle across a small skirmish-sized hex map. Units move over terrain that shapes movement and defense; players capture factories to reinforce and win by destroying the enemy's **flag unit**. Matches run **~10–15 minutes typical**, with the turn cap (§2.8) putting the **outer bound near 20 minutes** at a normal move pace against the AI — so 10–15 usual, ~20 worst-case if a match goes the distance to the cap. That's an expected duration, not a wall-clock guarantee (a PvP hotseat, a stretch feature, would add a per-turn timer). The pacing discipline is deliberate, see Lineage.

The deliverable is the game; AI agents are the method. The graded, shipped output is a complete, playable prototype — deliberately systems-forward and art-light so that AI agents can author, test, and tune the bulk of it, with the human acting as director. The genre was chosen for exactly this reason: a deterministic hex wargame is almost entirely **rules + data** — the surface agents author most reliably and can self-verify with tests. How much the agents actually did is reported honestly as a per-system provenance ledger (§3), as supporting evidence of the method — but the thing that ships and is judged is the game. (Accepted tradeoff: the art-light choice leaves visual-craft strengths on the bench, by design.)

Lineage. *Conflict* (Vic Tokai, NES, 1989) — the first NES game to render a hex battlefield. We inherit its structure (hex map, terrain modifiers, capturable cities/factories, destroy-the-flag win) and explicitly fix the flaw its own reviewers named: **Famicom Tsūshin (Famitsu) faulted it for battles that "took too long"** (Western outlets were kinder — Nintendo Power 4/5 — so the criticism is specifically about pacing, not the design). Everything here is tuned toward short, decisive play.

Design pillars. 1. **Legible, deterministic rules** — agent-authorable and unit-testable; no hidden RNG (seeded if any). 2. **Short and decisive** — small maps, punchy damage, a turn cap. Never an hour-long slog. 3. **Terrain + unit rock-paper-scissors** — the only depth needed; it comes from data, not feature count. 4. **Minimal art, maximal system.** 5. **Tests are a deliverable** — every core system ships with agent-generated tests and self-play balance data.

Scope at a glance. One polished scenario, four units, six terrains + the capturable Factory tile, capture + production, a heuristic AI opponent, and a functional UI is a *complete, shippable game.* Everything beyond that is garnish (see §2 scope table). Core playable in weeks 1–3; the remaining four weeks are content, balance, UI polish, and documenting the pipeline.

Success = a shipped, playable prototype. The primary bar is the game itself: the §2 core loop running, tuned, and fun in a 10–15 minute match. Agent contribution is tracked as a **per-system provenance ledger** (§3) — an honest, row-by-row record of which systems agents authored and verified — reported as supporting evidence of the method, *not* as the pass/fail bar. Anything a human hand-authored (UI polish, glue) is logged as human, so the ledger under-claims rather than over-claims.

Design decisions (all resolved). 1. ~~Scenarios in the prototype: one polished, or three?~~ → **RESOLVED: one polished scenario.** Content is the most agent-scalable axis; one hand-built map proves the pipeline without spending human review budget on volume. 2. ~~Economy: production only, or add the "fame" scoring spine?~~ → **RESOLVED (updated): a unified Fame currency.** Production points, combat rewards, and the win-score collapse into one **Fame** pool inherited from *Conflict* — earned from held factories and kills, spent to build, tallied to decide a capped match (§2.7). This is now core (not a separate stretch spine); flag-kill remains the primary, decisive win. 3. ~~Opponent: heuristic-only, or attempt the LLM commander?~~ → **RESOLVED: heuristic-only ships (§2.9); LLM commander stays a stretch toggle.** 4. ~~2-player hotseat: in scope or cut?~~ → **RESOLVED: stretch only, off the critical path (§2.10). Nearly free on IGOUGO, but doesn't serve the**

agent-authorship thesis, so it never blocks core. 5. ~~Working title.~~ → **RESOLVED: the game is titled *Stratocracy*.**

1.5 Revision Notes — First Draft → Final Draft

No external feedback was returned on the First Draft in time for this revision, so rather than wait, I **stress-tested the draft with an agent review crew** — the exact technique from the course's GDD-anatomy stress-test method (an Exploit-Hunter / Consistency / Pacing board run over the document, then human-adjudicated). Applying the course's own agent method to my own GDD *is* the growth here: the crew surfaced concrete, exploitable logic gaps on paper, and every one below was fixed before a line of code. Changes are listed as **finding → change → why it's better**.

#	Finding (agent review crew)	Change	Why it's better
1	**Turtle exploit in the tiebreak.** Original cap tiebreak ("most factories held, then total HP") *rewarded* stalling: grab one factory, wall the flag in a corner, run out the clock, win without fighting.	Reordered the cap tiebreak around **combat Fame earned** (damage dealt), excluding passive factory income; added a **mutual-passivity → draw** guard; put **multiple factories** on the map and a **live standings scoreboard** (§2.7–2.8, §2.11).	Turtling now *loses* the cap. Pillar 2 ("short & decisive") is enforced by the grading itself, not just by the turn cap.
2	**Success metric was unmeasurable.** "Agents did X% of the work" was a hero-number that couldn't be checked or defended.	Replaced it with a **per-system provenance ledger** — one row per system, marked by author and agent-verified status (§3).	It's *checkable* (re-runnable tests + commit trace) and deliberately under-claims (human work logged as human).
3	**Economy was unpriced.** Production points, combat rewards, and win-score were three separate uncosted pools; unit costs were undefined; built units had "nowhere to go / uncapped points."	Collapsed everything into one **Fame** currency with explicit costs (Inf 100 / Recon 150 / Arty 200 / Tank 300), factory +100/turn, kill ≈ ½ cost, flag +500 ends the match, and **space-throttled** spawning (§2.7).	One legible pool the player and the AI both reason about; the uncapped-points gap is closed by throttling on board space, not an arbitrary ceiling.
4	**The AI ignored the economy.** A re-review found the baseline opponent never built or captured — it would fight with only its starting force, making the whole economy one-sided.	Gave the baseline AI a two-phase turn: an **economy (build) phase** plus a **capture step** before the unit phase (§2.9).	Keeps production, capture, and the "objectives held" tiebreak live on *both* sides — the economy is actually exercised in play.
5	**Scoring could invert the win.** Raw-Fame scoring let a long cap match's accumulated kill-Fame "outscore" an actual flag kill, and a mutual-zero-combat cap had no defined result.	Made victory a set of **categorical tiers** (Decisive > Marginal > Draw); combat Fame is only the **sort key inside** the tiebreak, never the grade; mutual-zero-combat = **draw** (§2.8).	A flag kill is *always* the best outcome — the win condition can never be out-piled by attrition points.
6	**Framing risk (design judgment).** Positioning the *agent method* as the deliverable would have misaligned with a game-first rubric that grades the playable product first.	Reframed the identity: **the game is the deliverable**; agent-driven development is the *method*, reported as supporting evidence via the ledger (§1, §3).	Aligns the document with how the work is actually judged — the shippable game leads, the pipeline substantiates it.

1.6 Revision Notes — Final Draft → Production Draft

The Final Draft closed the *logic* gaps §1.5 found, but it left the document thin in four specific places that external feedback and the review board had already named. This revision closes them, and it was authored the same way the game is being built: a **four-author crew** — rules, UX/onboarding, scenario, and technical — writing in parallel against a frozen snapshot of this document, each confined to its own section and forbidden to edit the master. A separate **continuity gate** audited every draft against the live GDD for contradictions, stat drift, dead references, and invented numbers, and merge was blocked until it returned PASS. It did not pass on the first attempt: the gate filed four violations (a production-menu mock that greyed a unit the player could afford, a scoreboard chevron marking the wrong leader, and two dead references in the technical draft), the two authors responsible were re-spawned with those violations, and only the corrected drafts merged. That failure-and-correction loop is the same one §3 documents for T-COMBAT-07 — applied to prose instead of code. Changes are listed as **finding → change → why it's better**.

#	Finding (source)	Change	Why it's better
1	**No stated player-experience goals** *(rubric feedback, −0.5)*. The document said what the design *valued* (§1's pillars) but never what the player should *feel*, so nothing downstream could be checked against an intent.	Added **§2.0**, six goals PX-1..6, each carrying a rule source **and an observable check** — e.g. PX-4 "fighting is always better than hiding" is checked by T-CAP-02/03, "a side with zero combat Fame never wins a capped match."	A goal that cannot be checked is a preference, not a goal. Self-play reports and playtest notes now cite goals by ID, so intent is testable rather than asserted.
2	**No onboarding plan** *(review board open item)*. The GDD assumed a player who already understood hexes, Fame, capture, and the RPS triangle — with no manual and no tutorial mode in scope, nothing explained how they would learn them.	Added **§2.11.6**: three teachers (constraint, the forecast, and one-shot event tips), a scripted four-turn guided opening (turns 1–4) delivered through a directive strip, ten one-shot strings plus the two cap-approach banners, a pre-match briefing overlay, and a twelve-row concept ledger mapping every concept to where it is first taught.	The one genuinely unowned risk in a game with no manual is now a specified, buildable surface with a named cut line (§2.11.8) rather than an assumption.
3	**The ~3-match replay cliff** *(review board open item)*. The game is deterministic end-to-end, so any line that beats the AI once beats it forever; one mirrored map is a solved puzzle by match ~4.	Added **§2.13** in full — the shipped scenario *Ferrum Crossing* specified hex by hex, plus **§2.13.4**, which reframes the replay unit as a **configuration = map × seat × difficulty handicap** rather than as new mechanics.	Moves the cliff from match ~3 to match ~6 on shipped scope alone, using scenario data and one menu affordance — no new systems, no new rules, and the stretch maps stay explicitly cuttable.
4	**Eight `*pending*` ledger rows had no route to a gate** *(§3 provenance ledger)*. Combat cleared a real test gate; the	Added **§4.7**'s eight gateable spec stubs (Inputs / Formula / Invariants / Determinism / Acceptance, the shape that certified Combat)	Every pending row now has a named acceptance suite and a dependency position, so the ledger flips on evidence rather than on assertion — and

#	Finding (source)	Change	Why it's better
	other eight systems had no spec, no test IDs, and therefore no way to flip the same way.	**§4.8–§4.10** for data, integration and save-replay, and **§4.11**'s build order with an explicit critical path.	where a rule was missing, it is filed as a numbered open question in §4.7's register instead of being invented.
5	**§2 had drifted after the Conflict fold** *(continuity audit)*. Bridge and Factory had been folded into §2.3's terrain set, but §1 and §2.10 still said "five terrains," and §2.8's tiebreak had grown an apparatus nobody had re-justified.	Consolidated **§2.1–§2.10**, reconciled the terrain count across §1/§2.3/§2.10, and subjected the whole Fame/tiebreak apparatus to a **delete-test** — every piece removed in turn to see what breaks (§2.8's closing block).	The tiebreak now survives on evidence rather than inertia: the one piece that failed its delete-test (the floated per-turn Fame decay) was cut, and what remains is a guard, a sort, and an enum over state the game already tracks.

2. Game Mechanics

2.0 Player-experience goals

The pillars (§1) state what the design values; these goals state what the player should experience, phrased so each is observable in the test suite or the balance harness (§4.1). A goal that cannot be checked is a preference, not a goal — so each row carries its check. Self-play reports and playtest notes cite goals by ID.

ID	The player experiences	Rule source	Observable check
PX-1	Matches feel short and decisive: ~10–15 minutes typical, ~20 worst-case at the cap. A match never becomes the hour-long slog the lineage was criticized for.	§1, §2.8	Self-play turn-length distribution: the median match ends before the turn cap; cap-tiebreak matches are a minority of results.
PX-2	The rules can be trusted completely. The forecast shown before committing an attack is exactly what resolves — a whole turn can be planned in advance.	§2.6	Every attack's forecast equals its resolution; identical inputs give identical results (determinism invariants, T-COMBAT suite).
PX-3	Depth comes from where units stand, not from a memorized counter chart. The triangle is read off movement and range on the board.	§2.4	The type-effectiveness table ships all-1.0; no counter-multiplier is load-bearing at ship.
PX-4	Fighting is always better than hiding. There is no line of play where sealing a corner and running the clock wins.	§2.7, §2.8	A side with zero combat Fame never wins a capped match (T-CAP-02, T-CAP-03).
PX-5	The player always knows who is currently winning and how close the cap is. The tiebreak is never a hidden win condition.	§2.8, §2.11	The standings scoreboard displays every tiebreak input (combat Fame, objectives held *X/N*, surviving HP) plus the turn counter against the cap.
PX-6	The economy is one thought: earn Fame, spend Fame. The player never converts between currencies or tracks parallel pools.	§2.7	Every income, build cost, and combat reward mutates a single per-side Fame pool; no second resource exists in the data schema.

2.1 Core loop

```
Player turn:
  for each of your units (any order):
     select → move (within range, terrain-costed) → act (attack / capture
/ build) → done
  end turn
Opponent turn: same, driven by AI
repeat until a flag unit dies, an objective is met, or the turn cap
triggers a tiebreak
```

I-GO-U-GO alternation. No simultaneous resolution.

2.2 Hex grid

The battlefield is a field of pointy-top hexagons. The player reads adjacency and distance at a glance — every hex has six equal neighbours, so no direction is unfairly cheap the way diagonals are on a square grid — and each unit occupies exactly one hex. Line-of-sight blocking (a unit can't see or fire through a ridge) is a stretch goal.

2.3 Terrain (prototype set: 6 movement terrains + the capturable

Factory tile)

Terrain is the first thing the player weighs before moving: some hexes cost more of a unit's movement allowance to enter, some make a unit standing on them harder to kill, and some are closed to certain movement classes entirely — the player is always trading speed against cover and reach. Move cost is per movement class. All values are starter/tuning targets.

Terrain	Move cost	Defense	Passable
Plains	1	0%	land, air
Woods	2	+20%	land, air
Mountains	3	+40%	land (slow), air
Water	—	0%	sea, air
Town	1	+10%	land, air; capturable
Bridge	1	−10%	land, air; **the only hex a land unit crosses Water**
Factory	1	+15%	land, air; capturable; build/spawn + repair point

Bridge turns Water from dead space into a chokepoint system: land routes must funnel across bridges, and the negative defense makes a unit caught mid-crossing an exposed target — a contested kill-zone, not a safe shortcut. *(Tuning fallback: −10% → 0% if too punishing.)* **Factory** is the economy's anchor (§2.7): occupiable by either side, capturable by Infantry, a defensive anchor, and the build/spawn + repair point.

2.4 Units (prototype set: 4)

Rock-paper-scissors, not a roster. Starter stats.

Unit	HP	Move	Atk	Def	Range	Cost (Fame)	Notes
Infantry	10	3	4	2	1	100	Cheap; the only unit that captures towns/factories
Tank	20	5	8	5	1	300	Line breaker; **Flag Unit** is a Tank variant (not producible)
Artillery	8	3	10	1	2–3	200	Strong ranged, fragile, no melee counter
Recon/Air	12	7	5	3	1	150	High move; ignores some terrain cost

Cost is in Fame, the single currency (§2.7). All values are tuning targets, scaled down from *Conflict*'s cost ladder for four units and ~20-turn matches (§2.12).

Flag Unit: a designated Tank. Its death = loss. Not producible.

The triangle is positional, not a counter chart. Stratocracy's rock-paper-scissors lives in movement and range, an invariant worth naming: **Artillery beats Tank** (fires from range 2–3, takes no melee counter), **Recon beats Artillery** (move 7 runs it down into a fragile melee), **Tank beats Recon** (higher atk/def wins the range-1 fight). Infantry sits outside the triangle as the capture/objective unit. A type-effectiveness multiplier exists in the combat formula (§3) but ships **defaulted to 1.0 everywhere** — populated only if self-play shows the positional triangle too weak, so depth stays in positioning rather than in a lookup table (Pillar 3; PX-3).

2.5 Movement & pathfinding

The player selects a unit and every hex it can truly reach this turn lights up, terrain costs already accounted for — the highlight is the real move set, not an estimate. Clicking a lit hex sends the unit by the cheapest route. Only one unit fits in a hex, so lanes, chokepoints, and blocking are part of the plan. Zones of control — enemy units freezing anything that moves next to them — are cut from the prototype; they return only if matches feel too fluid.

2.6 Combat resolution

Before an attack is committed, the game shows the outcome up front: the predicted damage dealt, and the counterattack the defender strikes back with if it survives and the attacker is within its reach (attack forecast, §2.11). Combat is a pure function of attacker, defender, the defender's terrain, and the attacker's remaining health — the same matchup always resolves the same way, so a player can plan a whole turn and trust it plays out as shown (PX-2). Any randomness added later is **seeded**, so a given fight stays reproducible: the forecast the player sees is exactly what resolves, with no hidden roll.

2.7 Economy & capture — the Fame currency

Stratocracy runs on a **single currency, Fame** (inherited from *Conflict*, §2.12): earned from held objectives and combat, spent to build, tallied to decide a capped match. Production points, combat rewards, and the win-score are **one pool, not three** (PX-6). *All numbers are starter/tuning targets, scaled down from Conflict's for four units and ~20-turn matches.*

- **Factories & starting layout:** the map ships with **multiple factories** — a **home factory per side** (owned at start, so both players have income from turn 1) plus **two or more neutral factories** in contested ground; a typical small skirmish map has **~4 factories total**. The spread is what makes expansion worth fighting for and "objectives held" (§2.8) a real 0–N measure instead of a 1–0 coin-flip.
- **Income:** each **factory** held pays **+100 Fame/turn**; each captured **town** pays **+25/turn**. More objectives held = a faster army, so the neutral factories are the mid-game prize.
- **Build & spawn:** spend Fame at a factory to produce a unit from the buildlist (§2.4 costs: Infantry 100, Recon 150, Artillery 200, Tank 300). The unit **spawns on the factory hex if it's free, otherwise an adjacent free hex; if the factory is boxed in, the build waits.** Fame has no hard cap — deployment is throttled by board space, not a point ceiling.
- **Combat Fame:** destroying an enemy unit pays **~half its Fame cost** (e.g. a Tank kill = +150 — see Q5, §4.7, to make this exact); an **undamaged strike** (attacker takes no counter) pays a small bonus (see Q6, §4.7); destroying the enemy **flag pays +500 and ends the match**. These feed the same Fame pool and the cap tiebreak (§2.8).
- **Capture:** move an Infantry (the only capturer) onto a town/factory and hold to capture over N turns (start N=1–2); a captured objective flips its Fame income to the new owner.
- **Starting Fame:** each side opens with **200 Fame** (enough for two Infantry or one Artillery on turn 1) plus home-factory income from turn 1. Single-player difficulty is a starting-Fame handicap — see §2.9.
- **Repair:** a unit that ends its turn on an **owned Town or Factory and is not adjacent to any enemy** heals **+25% of max HP** (rounded down, min 1, capped at max) at the start of its next turn — free for the prototype. This is the game's only HP-recovery path; without it every unit is a one-way asset. The **not-adjacent clause is the anti-fortress lock**: a unit must break contact to repair, so a factory next to the front never becomes unkillable — and because repairing earns zero combat Fame, a repair-turtle still loses the cap tiebreak (§2.8). *(Heal %, and a possible small Fame cost, are tuning levers.)*

2.8 Turn structure & victory

- **Primary win — decisive victory:** destroy the enemy flag unit. Ends the match immediately.
- **Secondary win — territorial domination:** control **every factory on the map** at the start of your turn. Ends the match immediately and is ranked a **Decisive win**, equal to a flag kill. Because taking the last factory means capturing the enemy home factory deep in their territory, a flag kill is usually already available by then — this is an **active backstop that closes out a flag-turtle stalemate before the cap**, not a common win. Factories only (towns excluded), so it stays hard-won.
- **Loss:** your own flag unit is destroyed, or the enemy dominates all factories.

Turn cap → attrition tiebreak. If neither flag has fallen by the turn cap (**20 turns** on the shipped scenario — the cap is per-scenario data, set in the scenario file's `turnCap` field, §2.13.2 and §4.7 Stub 7), the match resolves as a battle of attrition. The full procedure is one guard, one three-key comparison, and one grade. Every input is a value the game already tracks for the economy (§2.7) and already shows on the standings scoreboard (§2.11): the tiebreak adds no new state, only an ordering over existing state.

Tally definition. A side's **combat Fame** is the Fame it earned from unit kills and undamaged-strike bonuses (§2.7). It **excludes** passive factory and town income — counting income would let a staller re-earn the turtle exploit (§1.5 #1) by sitting on objectives. The flag bonus (+500) can never appear in a capped tally: destroying the flag ends the match immediately, so no match that reaches the cap contains one.

Resolution procedure, in order:

1. **Mutual-passivity guard.** If *both* sides' combat Fame is zero — nobody engaged — the match is an immediate **draw**. It does not fall through to the keys below, because "objectives held" would otherwise re-crown a turtle who simply sat on more factories.
2. **Lexicographic comparison** — higher wins at the first key that differs:
    1. **Combat Fame earned.** The anti-turtle lever (PX-4): a player who sealed a corner and refused to fight scores zero here and loses the cap to anyone who dealt damage. (The guard already covers the both-zero case, so no separate "both sides fought" precondition is needed on the later keys — if this key ties at a nonzero value, both sides fought by construction.)
    2. **Objectives held** — the factories and captured towns a side owns at the cap, as *X of N*. Ownership only: a capture in progress (§2.7) counts for nobody until the objective flips. Towns count here even though the domination win is factories-only — domination must stay hard-won; this key merely breaks an exact Fame tie. With ~4 factories plus towns on the map (§2.3, §2.7) this is a genuine spread.
    3. **Surviving strength** — total remaining HP of a side's units. Deliberately last: alone it would re-reward turtling, but by the time it applies both sides have earned identical nonzero combat Fame, so an HP edge measures trading efficiency between two sides that both fought. It is free to compute (the scoreboard already sums it) and decides matches that would otherwise be draws.
3. All three keys equal → **draw**. Acceptable for the prototype; a competitive build would add a sudden-death overtime.

Victory quality. A result is graded as a **tier**, not a number:

Tier	Trigger
Decisive win	Enemy flag destroyed, or territorial domination
Marginal win	Led the attrition comparison at the cap
Draw	The passivity guard fired, or all three keys tied

Tiers rank categorically: a **decisive win always outranks a marginal win**, regardless of how much Fame either side piled up. Combat Fame is only the sort key *inside* the tiebreak, never the grade — keeping the two separate prevents a long capped match's accumulated kill-Fame from "outscoring" an actual flag kill (§1.5 #5). For self-play tuning, log the tier plus the Fame breakdown per match. The result: a flag kill is always the best outcome, so Pillar 2 ("short and decisive") is enforced by the grading itself, not just by the turn cap.

Invariants (each phrases directly as a `T-CAP-` test):

1. **T-CAP-01** — Flag destruction on any turn at or before the cap yields a Decisive result; the tiebreak procedure is never evaluated.
2. **T-CAP-02** — Both sides at zero combat Fame at the cap → Draw, regardless of objectives held or surviving HP.
3. **T-CAP-03** — Passive income never decides the cap: a side holding 4 factories with zero kills loses the cap to a side holding 1 factory with one Infantry kill.
4. **T-CAP-04** — No capped tally contains the +500 flag bonus.
5. **T-CAP-05** — A capture in progress at the cap contributes zero to "objectives held" for either side.
6. **T-CAP-06** — Tier order is Decisive > Marginal > Draw for *any* pair of Fame totals.
7. **T-CAP-07** — Determinism: identical end state at the cap → identical result and tier.
8. **T-CAP-08** — Controlling every factory at the start of your turn ends the match immediately as a Decisive win; towns do not count toward domination.

> **Why this shape (the delete-test).** Every piece of the apparatus was tested by deletion. Delete key 1 → the documented turtle exploit (§1.5 #1) returns whole. Delete the guard → a four-factory turtle beats a one-factory turtle without a shot fired. Delete key 2 → contesting the neutral factories (§2.7's mid-game prize) stops mattering at the cap. Delete key 3 → more draws for zero savings, since the HP sum already exists for the scoreboard. Delete the tiers → a capped grind's tally can read as "beating" a flag kill in tuning logs — the inversion §1.5 #5 closed. Delete the domination backstop → a walled-in flag forces every such match to run the full cap for a Marginal result instead of ending early and Decisively (Pillar 2). The one piece that failed the test — the floated per-turn Fame decay — is cut in this revision (§1.6 row 5). What remains is a guard, a sort, and an enum, all over state the game tracks anyway.

On the turn cap vs. real time. The cap bounds *turns*, which guarantees the match terminates; it does not bound wall-clock minutes. Against the shipping single-player AI (which moves instantly) that is a non-issue, so "~20 minutes" (§1) is an expected duration at a normal move pace, not a hard real-time ceiling. The only mode that would need a per-turn timer is **PvP hotseat**, a stretch feature (§2.10) — if it ships, add a move clock there.

2.9 Opponent AI (runtime gameplay system)

- **Baseline (ships): simple objective-seeker.** Two phases each turn, so the AI actually *uses* the economy (§2.7), not just the map:
  - **Economy phase.** At each factory it holds: if it can afford a unit, build one from a default buildlist (mostly Infantry, an occasional Tank), spawning per §2.7. It spends Fame and replaces losses instead of hoarding.
  - **Unit phase, per unit:** (1) an idle **Infantry** adjacent to or near an uncaptured, **undefended** factory/town moves onto it to capture — the AI contests objectives, keeping capture, production, and the "objectives held" tiebreak live on *both* sides; (2) if an enemy is within reach after moving, attack — prefer the enemy flag, else the best expected-damage target; (3) ranged units (Artillery) fire from maximum standoff so they don't eat a counter; (4) otherwise advance along the cheapest path toward the enemy flag.
  - One cheap guard keeps it from looking broken: **skip a strictly-losing attack** (the unit would die and trade down).

  Deliberately un-clever — decisive, readable, and fully testable. This is the shipping opponent.
- **Difficulty = a starting-Fame handicap, not a smarter AI** (mirrors *Conflict*). The baseline routine is identical at every tier; only the economy shifts: **Easy** = player +150 opening Fame; **Normal** = even (200/200, §2.7); **Hard** = player −100. Deterministic and trivially tunable, with no AI-quality risk. *(Stretch dial, documented only: also scale AI income ±25%/tier.)*
- **Stretch: AI second pass (utility + threat map).** Upgrade the baseline to score candidate actions on attack value (expected damage vs. counter), objective proximity, and safety (a threat map of enemy reach). Week-3 work, only if the baseline proves too exploitable in self-play.
- **Stretch: LLM commander.** Serialize the board to text, enumerate legal moves, ask the model to rank one, then **validate and apply** (never trust an unchecked move). Behind a toggle, heuristic as fallback. This is the one place an AI *agent* appears in the shipped product — see §3.

2.10 Scope table

	Contents
IN (core; phased wk 1–3 per §4.4)	Grid; 4 units; **6 terrains + the Factory tile**; move + attack; **multiple factories** (home-per-side + contested neutrals, ~4) + capture + Fame production *(these land wk 3, not wk 1–2)*; heuristic AI; **one hand-built scenario**; win-by-flag; functional UI
STRETCH (if ahead)	2nd–3rd scenario; LLM commander; fog/recon; sea/air units; map-gen MCP toolset; 2-player hotseat
CUT	All 16 original scenarios; campaign/meta; zones of control; elaborate art; anything real-time

2.11 UI/UX

Stratocracy's interface has one job: make a deterministic game *look* deterministic. Every rule in §2 is knowable before the player commits, so the UI's standard is not "pretty" but **no surprises** — if the player is ever surprised by an outcome, the UI failed, not the player. Three principles govern everything below:

1. **Hover is information, click is commitment.** Hovering never changes game state; it only reveals (terrain lines, path previews, the forecast). A single left-click commits. Cancel is always right-click or Esc.
2. **Every element earns its pixels.** Each HUD element is audited against the decision it supports (§2.11.2); anything supporting no decision is cut. Two deliberate cuts up front: **no minimap** (the one shipped scenario is a small skirmish map, §2.7 — it fits on screen at default zoom; pan/zoom covers the rest) and **no combat log** (determinism plus the forecast make a history redundant for the prototype; a log rides the replay format if that ships, §4.10).

3. **Rules-critical information is never transient-only.** Toasts and banners are receipts; the durable version of every fact lives on a persistent or hover-recallable surface.

UI is budgeted as core work from week 2 (§4.4, §4.5 — "UI underestimated" is a named risk) and everything below is scoped to what one person can build in UMG on top of agent-scaffolded widget skeletons (§3, UI Scaffolder role).

2.11.1 Control scheme

Selection state machine. The core loop (§2.1: select → move → act → done, any unit order) maps to four UI states:

```
IDLE ——LMB on own unacted unit——▶ SELECTED (reachable hexes lit, §2.5)
IDLE ——LMB on own factory——————▶ PRODUCTION MENU (§2.11.5)
SELECTED ——LMB on lit hex——————▶ MOVED (attack targets lit; Wait
available)
SELECTED ——RMB / Esc———————————▶ IDLE (nothing committed)
MOVED ——hover enemy target—————▶ forecast card shown (§2.11.3)
MOVED ——LMB on lit target——————▶ attack resolves exactly as forecast →
unit DONE
MOVED ——Space (Wait)———————————▶ unit DONE without acting
MOVED ——RMB / Esc——————————————▶ unit DONE where it stands (move already
spent)*
```

* Under a ruling that grants move-undo (**Q11**, §4.7 — currently unruled, and no gate assumes it), RMB/Esc in MOVED instead reverts the unit to its pre-move hex and returns to IDLE. Until that rule is adjudicated, the shipping semantics are the conservative ones shown: a completed move stands. The two behaviors share a UI; only the rules module differs.

Capture and build need no extra verbs. Capture is by presence (§2.7: an Infantry that ends its move on a capturable tile begins capturing — a progress pip appears, no button). Building is the factory's own interaction, not a unit's. This keeps the per-unit action vocabulary to exactly two: *attack* or *wait*.

Input reference.

Input	Effect
Hover hex	Terrain line in the info panel (§2.11.2); if a unit is SELECTED, dotted **path preview** along the cheapest route (§2.5) with the terrain-cost tick per hex
Hover unit	Unit stats in the info panel; if own MOVED unit has it in reach, the **forecast card** (§2.11.3)
LMB	Select own unit / commit previewed move / commit forecast attack / open production menu on own factory / activate buttons
RMB / Esc	Cancel: close menu or forecast, deselect, back out one state (see machine above)
MMB drag or **WASD / arrows**	Pan camera
Mouse wheel	Zoom (two or three fixed steps, not continuous — readability over cinematography)
Tab	Cycle to the next unit that has not acted
Space	Wait — mark the selected/moved unit done without acting
F	Snap camera to your flag unit
B	Open the production menu at the selected owned factory (or your home factory if none selected)
Enter	End turn — with a confirm dialog if any unit has not acted: `3 units have not acted. End turn?`
Z	Undo move *(only if Q11 (§4.7) is ruled to grant undo; otherwise unbound)*
Any click / Esc during enemy turn	Skip AI playback to its end state (§2.11.2, turn banner)

No hidden double-functions, no drag-to-move, no context menus. A first-session player can complete a match knowing only: hover, LMB, RMB, Enter.

2.11.2 HUD layout & information architecture

Screen layout.

```
+----------------------------------------------------------------------
-+
| +-------------------+   [ DIRECTIVE STRIP / TURN BANNER ]   +----------+
|
| | TURN 12 / 20      |        (top center, transient)        | FAME 350 |
|
| |-------------------|                                       | +175/turn|
|
| |        YOU  ENEMY |                                       +----------+
|
| | Destr. 450   600◂ |
|
| | Obj.   4/8   3/8  |
|
| | HP     47    55   |          ( hex battlefield )
|
| +-------------------+
|
|   (live scoreboard,               [H] flag markers on-map
|
|    §2.11.4)                       [forecast card floats near target]
|
|
|
| +-------------------+                        +--------------------+
|
| | INFO PANEL        |      [ toast queue,          | 3 units idle      |
|
| | Woods             |        bottom center ]       | [ END TURN  ⏎ ]
| |
| | move 2 · def +20% |                        +--------------------+
|
| +-------------------+
|
+----------------------------------------------------------------------
-+
```

Three information layers, strictly tiered:

1. **Persistent** (always on screen): scoreboard + turn counter, Fame pool + income rate, End Turn + idle-unit count, on-map flag `H` markers and unacted-unit pips. These are the four standing decisions of every turn — *am I winning the cap, can I afford to build, is my turn actually finished, what must I protect.*
2. **Contextual** (selection-driven): reachable-hex highlight, path preview, attack-target highlight, info panel content, forecast card, production menu, capture pips. Appears with a selection or hover, gone when it ends.
3. **Transient** (event receipts): toasts (`+100 Fame — Factory`, `+[N] HP — repaired`, `+150 Fame` on a kill), turn banner, one-shot onboarding lines (§2.11.6), cap-approach banners (§2.11.4). Each transient fact has a durable home: income toasts restate what the Fame widget's `+X/turn` already shows; kill toasts restate the scoreboard's Destroyed row.

Earn-your-pixels audit — every persistent/contextual element, the decision it supports, and the rule it surfaces:

Element	Decision it supports	Rule surfaced
Scoreboard (turn + 3 rows)	Force combat or hold; how close is the cap	§2.8 tiebreak order, turn cap
Fame pool + `+X/turn`	Build now vs. save; which neutral factory is worth a fight	§2.7 income (+100 factory / +25 town), costs
End Turn + idle count	Is my turn genuinely spent	§2.1 per-unit loop
Flag `H` marker (both sides, always visible)	What to protect, what to hunt	§2.4 flag death ends the match; *Conflict*'s `H` convention
Unacted pip on own units	Which units still have a move	§2.1
Reachable-hex highlight	Where can this unit truly go	§2.5 — "the real move set, not an estimate"
Path preview with cost ticks	Which route; exposure en route (e.g. a turn spent on the Bridge at −10%)	§2.3, §2.5 cheapest-route
Attack-target highlight (incl. Artillery's dark range-1 hole)	Who is actually hittable from here	§2.4 ranges; the Artillery dead zone
Forecast card	Commit this attack or not	§2.6 — the whole section (§2.11.3)
Info panel	Speed-vs-cover terrain trade; matchup reading	§2.3 table, §2.4 stats
Production menu	What to buy, where it will spawn	§2.7 build & spawn rules (§2.11.5)
Capture progress pip	Hold or abandon the capture	§2.7 capture over N turns
Repair eligibility pip (owned tile, damaged unit selected, no adjacent enemy)	Retreat to heal or keep fighting	§2.7 repair + anti-fortress clause

Info panel (bottom-left, hover-driven, ~3 lines, never modal): - Hovered hex: terrain name, move cost, defense bonus, and status if capturable — `Factory · move 1 · def +15% · yours (+100/turn)` or `· neutral` or `· enemy`. - Hovered unit: name, HP as `12/20`, Atk/Def/Move/Range, `has acted` flag. The flag unit's panel is red-edged and appends `FLAG — its loss ends the match.` - Empty when nothing is hovered. It never covers the board's lower-center where fighting happens.

Turn banner & AI playback. A brief `YOUR TURN` / `ENEMY TURN` banner marks the IGOUGO handoff (§2.1). The headless AI resolves instantly (§2.8); the presentation layer **replays its action list at a watchable fixed pace** (~0.5 s per action, camera stepping to each) so the player can read what the AI built, captured, and attacked — this is presentation pacing only, no rules change. Any click or Esc skips to the end state. First-session value: watching the AI's economy phase is how the player learns the enemy shares the same Fame economy (§2.9).

2.11.3 The attack forecast — the game's centerpiece display

Combat is a pure function (§2.6, §3 spec), so the forecast is not an estimate — it is the resolution, shown early. It appears on **hover** over any lit target from the MOVED state; **LMB commits**, RMB/Esc cancels. The card:

```
+-----------------------------------------+
| ATTACK FORECAST                         |
| Artillery  →  Tank      (Woods +20%)    |
|                                         |
|  You deal    3 dmg      Tank 20 → 17    |
|  Counter     0          out of range    |
|                                         |
|  [LMB] Commit        [RMB / Esc] Cancel |
+-----------------------------------------+
```

(Values follow the §3 formula: Artillery atk 10 at full HP vs Tank def 5 in Woods → round(10 × 1.0 × 0.8) − 5 = 3; counter 0 because the attacker sits outside the Tank's range 1.)

Hard rules for the card:

- **The defender's terrain bonus is named inline** (`Woods +20%`), every time — terrain defense must never read as hidden dice. (Per the §3 invariant, it is always the

defender's hex; the card's placement of the modifier next to the defender teaches that for free.)
- **The counter line is never omitted, and always states its reason**: a number, `out of range`, or `defender destroyed`. This line is where the positional RPS triangle (§2.4) becomes visible — there is no type chart to consult, deliberately, so `Counter 0 — out of range` *is* the Artillery-beats-Tank lesson and `Counter 5` on an adjacent brawl *is* the Tank-beats-Recon lesson.
- **HP is shown before → after** for the defender (and for the attacker whenever the counter is nonzero), so the HP-scaling term of the formula (§2.6 — a damaged attacker hits softer) is observable across fights rather than asserted.
- **Lethal forecasts state their reward**: if the defender dies, the card appends `Destroys Tank · +150 Fame` (kill ≈ half cost, §2.7) — the tiebreak's combat-Fame criterion (§2.8) is priced on the commit card, not discovered at the cap.
- **Flag warning band**: if the forecast is lethal to *either* flag, the card gains a band — `FLAG AT RISK — this attack ends the match` (own flag: red; enemy flag: gold, with `+500 · Decisive victory`). No player can end a match, theirs or the enemy's, without having been told on the card they clicked.
- **Determinism is restated once**, via the first-attack one-shot (§2.11.6): `Check the forecast. It is exact — what you see is what resolves.` After that the card carries no reassurance text; the outcomes matching the card, every time, are the proof.

Selecting Artillery renders its attack ring as range 2–3 **with the range-1 hole drawn visibly dark** — the dead zone on the map is the "Recon runs it down" lesson in pixels, using the attack-target highlight already budgeted.

2.11.4 The live Fame scoreboard

The scoreboard exists because of revision §1.5-#1: the tiebreak must never be a hidden win condition. It is persistent (top-left), compact, and its **rows are ordered top-to-bottom in exact tiebreak order (§2.8)** — the layout *is* the rule, read passively all match:

```
+--------------------------+
| TURN 12 / 20             |
|--------------------------|
|            YOU      ENEMY |
| Destroyed  450      600 ◂ |
| Objectives 4/8      3/8   |
| Unit HP    47       55    |
+--------------------------+
```

- **Turn counter** against the cap, always. (`/ 20` is *Ferrum Crossing*'s cap, §2.13.2; the cap is per-scenario data, so the widget reads `turnCap` from the scenario rather than hardcoding a number.)
- **Destroyed** = combat Fame earned (kills + undamaged-strike and flag bonuses, §2.7) — **passive income is excluded**, exactly as the tiebreak excludes it. Hover tooltip: `Fame from kills and strike bonuses. Factory income does not count at the cap.` This row deliberately does *not* equal the spendable Fame pool (top-right widget), and the tooltip on each names the difference — the one place the single-currency design (§2.7) needs a disambiguating sentence.
- **Objectives** as *X of N* over all factories + capturable towns (§2.8 criterion 2), N supplied by the scenario (§2.13) — **N = 8** on *Ferrum Crossing* (4 factories + 4 towns), as the mock shows.
- **Unit HP** = surviving strength (criterion 3), listed last because it *is* last.
- A **chevron (◂)** marks the current attrition-tiebreak leader, evaluated in criteria order, and flips visibly when the lead changes. It is drawn beside the leading side's value — in the mock above the enemy leads at criterion 1, 600 combat Fame to 450 (§2.8: higher wins), so the chevron sits on the enemy column. If both Destroyed values are zero, the chevron is replaced by `— no engagements —` spanning the row: the mutual-passivity draw (§2.8) made visible before it bites.
- **Cap-approach banners** (transient, once each): at cap−5, `5 turns to the cap. The scoreboard decides a capped match.`; additionally, if both sides are still at zero combat Fame, `No engagements. A capped match with no combat is a draw.`

End-of-match screen. The result is the tier first (§2.8 — Decisive / Marginal / Draw), then the same three rows in the same order, so the verdict is always a restatement of what was on screen all match. Beneath the tier, one **faction-voiced result line** (per the setting guide: faction voice appears only on result screens; ≤ 30 words; field-manual register). Samples, one per case, generated content to follow these:

- Directorate, decisive: `Command directive fulfilled. The enemy flag is struck from the record. Order is restored.`
- Directorate, marginal: `The cap is reached. The ledger favors the Directorate. The record stands.`

- Vanguard, decisive: `Their flag is down. We hold the ground. That's the whole report.`
- Vanguard, marginal: `Cap hit. We did the damage; they held the rear. The ground says we win.`
- Draw, neutral system voice: `Turn cap reached. Attrition equal. Recorded as a draw.` / mutual passivity: `Turn cap reached. Neither side engaged. Recorded as a draw.`

2.11.5 Production menu & match-flow surfaces

Production menu — opens on LMB on an own factory (or `B`), anchored beside it:

```
+-------------------------------------+
| FACTORY — BUILD          Fame: 250   |
|-------------------------------------|
|  Infantry   100   [Build]            |
|  Recon      150   [Build]            |
|  Artillery  200   [Build]            |
|  Tank       300    ----   (need 50)  |
|-------------------------------------|
|  Spawns here, or adjacent if         |
|  occupied.                           |
+-------------------------------------+
```

- Unaffordable rows are greyed with the shortfall named (`need 50` at the mocked 250-Fame pool against the Tank's 300 cost, §2.4), never hidden — the price list is also the strategy lesson (§2.4 costs).
- The spawn rule (§2.7: factory hex if free, else an adjacent free hex) is one static line in the footer. If the factory is fully boxed in, the footer swaps to `Boxed in — build waits for a free hex.` and Build buttons disable: the space-throttle (§2.7) explains itself at the moment it applies.
- When any unit is affordable and the factory has not built this turn, the factory tile shows a small `BUILD` pulse — the nudge that connects hoarded Fame to an army (a first-session failure mode, §2.11.6 ledger).
- The row hover shows the unit's stat line in the info panel, so buying is done with the §2.4 table in view.

Pre-match: the static briefing overlay (three callouts: flag, Bridge, factories — §2.11.6-A). **Post-match**: the end screen above. That is the complete screen list for the prototype: title/menu, briefing, match, result. No settings screen beyond volume + resolution is budgeted (Enhanced Input remap is a polish item).

2.11.6 Onboarding — the first session

Philosophy. No manual, no tutorial mode, no modal text walls. Three teachers:

1. **Constraint** — the first turn removes every option except the one being taught.
2. **The forecast** (§2.11.3) — deterministic combat means every attack is a free, truthful lesson in terrain defense, range, and HP scaling. The plan makes the player *read* the forecast once, early; after that the game teaches itself.
3. **One-shot event tips** — a single system-voice line (≤ 30 words, tone bible), fired the first time a concept becomes relevant, never repeated (boolean flags in the save slot, §4.1).

The first match runs on the one shipped scenario at **Easy** by default (player +150 opening Fame, §2.9) with a **guided opening**: four scripted directives inside a fixed four-turn window — the first appears on turn 1, the strip and every beat behind it are gone for good at the end of turn 4, then hands-off. Any completed match on the save skips all guidance automatically; a `Skip guidance` control kills it instantly for anyone, and kills the guided opening's board state with it — the objective ring and the turn-1a unit marker clear in the same frame as the strip.

A. Pre-match briefing — a dimmed board, three anchored callouts, click-through (~5 s): **your flag** (`If it falls, you lose. It cannot be rebuilt.`), **the Bridge** (`The only land crossing.`), **factories** (`Hold them for Fame. Fame builds units.`). Why only three: the flag is the win condition, factories are the economy, and the Bridge is the map-reading trap — a player who misses the single Water crossing misplans the whole match.

B. Guided opening (turns 1–4), via a one-line directive strip at top center, one instruction at a time, each retiring on completion:

Turn	Constraint	Directive	Teaches	Retires when

Turn	Constraint	Directive	Teaches	Retires when
1a	Only one marked Infantry selectable, others dimmed (hover: `Locked this turn.`). End Turn is inert until that Infantry has moved (hover: `Move the marked Infantry first.`) — this is what makes 1a retire inside turn 1 in every branch, and it is the only guided-opening constraint that gates a player *input* rather than a selection, so it is carried as **pending Q27** (§4.7) rather than adopted silently	`Select the marked Infantry. Lit hexes are its true reach. Click one to move.`	Selection; the highlight is the real move set (§2.5)	Move completes
1b	End Turn pulses	`End turn. The enemy moves; then you.`	IGOUGO (§2.1) — the player watches a full AI turn	Enemy turn ends
2 *(standing)*	None on selection. The scenario's designated neutral factory (`guidedOpening.objective`, §2.13.1) is ringed from turn 1; its info-panel line appends `Only Infantry captures.`	`Move the Infantry onto the ringed Factory. Only Infantry captures.`	Capture; the Infantry-only rule (§2.7)	A capture pip appears — on whatever turn that happens. *Standing* means it stays **outstanding**, not that it holds the strip. How many turns it actually holds the line is decided by rules 1–2 and by when the pip lands, and the schedule table's three branches are exhaustive: **twice** — turn 2, then a turn-4 rule-2 last call, if the pip has still not landed (wandered); **once** — turn 2 only, retiring on a turn-2 or turn-3 pip; or **never** — the pip lands on turn 1 and beat 2 retires before rule 1 can select it (fast lane). On every turn it does not hold the line it runs on the ring and the unit marker. Hard-expires at end of turn 4
3	None. Fame ≥ 100 guaranteed by 200 start +	`Spend Fame at your Factory.`	Fame → factory →	A unit spawns — on whatever turn that happens

Turn	Constraint	Directive	Teaches	Retires when
	home income (§2.7)	`Infantry costs 100.`	unit	that happens, including turn 1

The Turn column is the beat's **order index** — the turn it takes the line in the common case — not a floor. Rules 1–2 below assign the line, so a beat moves up whenever a lower-numbered beat has already retired: beat 3 takes turn 2 in the fast-lane branch of the schedule table. The strip shows **one directive at a time**, and the line is assigned at the start of each turn by two rules in order:

1. the lowest-numbered **outstanding** beat that has **not yet held the line on an earlier turn**;
2. if every outstanding beat has already had its turn on the line, the lowest-numbered outstanding beat — a **last call**.

A beat gives up the line either the instant it retires (which is how 1a hands off to 1b inside turn 1) or at the end of the turn it first appeared, whichever comes first. **Giving up the line is not retiring.** An outstanding beat that has yielded keeps running on its board state: for beat 2 that is the ringed objective and the marked Infantry, which are the same instruction in spatial form and have been on screen since turn 1. The text line introduces the objective; the ring is what reminds.

The consequence that matters: **no beat expires unheard.** Each beat either retires on its own event — its lesson landed in play, which is exactly what beat 2 does on turn 1 in the fast-lane branch, before it has ever held the line — or it holds the line at least once before the window closes, which is what rule 2 exists to force. A beat 2 that hangs — the *Ferrum Crossing* 1-MP-slack case below — cannot starve beat 3, whose Fame → factory → unit lesson is the one the §2.11.6-D ledger confirms with a bought unit on the board. Rule 2 then makes turn 4 a last call for whatever is still outstanding, so the strip's final turn is spent on the player's actual gap rather than on order-of-arrival:

Turn	Common case — pip lands turn 2	Wandered case — pip lands turn 3 or 4	Fast lane — pip lands turn 1 (*The Causeway*, 3 MP lanes)
1	1a, then 1b when 1a retires	1a, then 1b	1a, then 1b; the pip lands inside this turn, so beat 2 retires **without ever holding the line**
2	beat 2 (rule 1) — retires on the pip	beat 2 (rule 1) — holds, then yields	beat 3 (rule 1)
3	beat 3 (rule 1)	beat 3 (rule 1)	beat 3 — rule 2 last call, **untagged**
4	beat 3 — rule 2 last call, tagged	beat 2 — rule 2 last call, tagged	beat 3 — rule 2 last call **again**, tagged
end of 4	strip gone; anything still outstanding expires	strip gone; anything still outstanding expires	strip gone; anything still outstanding expires

How to read a cell. Each cell names the beat that rules 1–2 select for that turn, given the column's pip timing *and* beat 3 still outstanding when its cell is reached. Beat 3 retires on a spawn **event**, not on turn 3, and Fame ≥ 100 from turn 1, so a player who builds early retires it early. Every cell is therefore read as: *this beat — or, if it has already retired, the next beat rules 1–2 select — or nothing, if none is outstanding.* Turn 1 is identical in all three columns because 1a and 1b cannot outlive it: End Turn stays inert until the marked Infantry moves (beat table, row 1a), and 1b's retire condition **is** the turn boundary.

Is the strip ever quiet before the end of turn 4? No — there is no live-but-blank strip in this system. Rule 2 has no exit: while any beat is outstanding, some beat holds the line. The only empty state reachable before end of turn 4 is the strip being **gone**, and its condition is exact — *all four beats retired*: 1a and 1b inside turn 1, beat 2 on its pip, beat 3 on a spawn. The earliest that can occur is the end of turn 1, and only in the fast-lane branch with a turn-1 build; it is permanent, per the disappearance rule below, not a pause. In particular, fast-lane turn 4 is **not** unconditionally quiet: a player who has not bought a unit still has beat 3 outstanding, and rule 2 returns it to the line, exactly as the common-case column does at its own turn 4. That is the guarantee working, not an exception to it.

A **turn-4** last-call line is the beat's own text with a dim right-hand tag, so the player is never dropped mid-instruction without warning. The tag states a fact about the *window*, not about rule 2, so it renders on turn 4 only: an earlier rule-2 last call (fast lane, turn 3) shows the same line untagged, because guidance does not end that turn and the strip must not say it does:

```
+-----------------------------------------------------------------------
-+
|  Move the Infantry onto the ringed Factory. Only Infantry captures.
|
|                                                        guidance ends this
turn|
+-----------------------------------------------------------------------
-+
```

The strip disappears for good once all four beats have retired, and unconditionally at the end of turn 4; every beat, the objective ring, and the turn-1a marker expire with it.

Why beat 2 is a standing directive, not a turn-2 deadline. Its target is guaranteed by §2.13.1's **opening-capture reachability** invariant: for each seat, at least one Infantry deployment hex has a Bridge-free land path to a **neutral factory** costing ≤ 6 movement points — two turns at Infantry Move 3 (§2.4) — and the scenario file names that unit and that factory in `guidedOpening.infantry` and `guidedOpening.objective`. The directive strip reads exactly those two fields: `guidedOpening.infantry` is the Infantry marked in beat 1a, `guidedOpening.objective` is the factory ringed from turn 1. Nothing is measured at runtime and no "nearest objective" heuristic is used — the lane is authored, machine-validated, and recorded as a number by `validate_scenario` (§4.2), so the onboarding and the map can never disagree about which factory the player was told to take.

What the invariant does *not* buy is a safe turn-2 deadline, and that is why beat 2 retires on an event rather than a turn number. Beat 1a hands the player a free move in any direction — that is its whole lesson, and the onboarding must not punish the player for using it. On *Ferrum Crossing* both lanes cost **5 movement points**, not 5 and 4 hexes: West (1,5) → South (5,7) is 5 hexes of Plains at cost 1, and East (9,3) → North (6,2) is 4 hexes but one is a mandatory Woods ring hex at cost 2 (§2.3). Against the 6 MP budget that is **1 MP of slack**, the tightest of the three maps and the only one that is tight at all: both stretch maps were redrawn on even row counts (*Longwater March* 13 × 8, *The Causeway* 9 × 8, both `rot180`), and their lanes price at **4 MP / 4 MP** and **3 MP / 3 MP**, so they carry **2** and **3** MP of slack respectively (§2.13.1's table). No lane on either 8-row map carries 4. A single turn-1 step spent walking off the lane therefore pushes the pip to turn 3 — and a hard turn-2 retire condition would strand the strip on a directive the player had already been made unable to satisfy that turn. The standing directive absorbs precisely that: on *Ferrum Crossing*, where a turn-1 pip is impossible at 5 MP per lane against Move 3, it takes the line on turn 2, stays outstanding until the pip appears, and hard-expires at the end of turn 4 — the last turn of the guided window, not one turn past it. It persists as an *objective*, not as a line of text: it holds the strip for turn 2, yields the line to beat 3 for turn 3, and returns for a turn-4 last call only if the pip still has not landed. That is **two turns on the line at the outside** — one in the common case, where the turn-2 pip retires it, and none at all on a map whose lanes let the pip land on turn 1 (the fast-lane column). That is the whole price of "standing", and it is paid in ring and marker rather than in strip time, so the Fame → factory → unit lesson is never the thing that gets crowded out by a slow walk. Ringing the objective from turn 1 biases beat 1a's free move onto the lane without constraining it, so in the common case the slack is never spent and the pip lands on turn 2 as designed.

"Capturing" by turn 2, never "captured." The invariant promises the Infantry is *standing on* the factory at the end of turn 2, not that the tile is yours: under Q4's N = 1 reading the tile flips at the start of turn 3 (§2.7). So the directive reads `Move the Infantry onto the ringed Factory` — never "capture it" — and it retires on the **capture pip**, the arrival receipt, not on the ownership flip. The flip gets its own confirmation one turn later via the `+100 Fame — Factory` toast, which is where the concept ledger's Capture row already ends. This wording is also correct if N is ever ruled 2: the pip is the arrival event either way.

C. Event-driven one-shots (once each, at first relevance): first attack hover → `Check the forecast. It is exact — what you see is what resolves.`; first Artillery strike at range 2–3 → `No counter at this range. Artillery strikes beyond reply.`; first Recon-vs-Artillery forecast → `Recon closes fast. Artillery cannot answer at range 1.`; first Tank-vs-Recon forecast at range 1 → `Armor wins the adjacent fight.`; first Water hover with a unit selected → `Impassable to land. Cross at the Bridge.`; first Bridge hover → `The only crossing. -10% defense — do not stall on it.`; first combat Fame → `+150 Fame. Kills count at the cap. Income does not.` *(amount = actual award)*; first undamaged strike → `+[N] Fame — undamaged strike.`; first repair blocked by adjacency → `No repair — enemy adjacent. Break contact to repair.`; first repair tick → `+[N] HP — repaired at Factory.`; plus the two cap-approach banners (§2.11.4).

D. Concept ledger — every concept, its failure point, its teach moment, its confirmation:

Concept	Where a first-timer gets lost	Taught at	Confirmed landed when
Hexes & movement	Clicks the map with nothing selected	Turn 1a: single selectable unit + reach highlight	Move completes; directive retires
IGOUGO turns	Expects simultaneous movement	Turn 1b: pulsing End Turn, then watching the AI's turn	Player ends turn 2 unprompted
Terrain move cost	Lopsided highlight (shallow into Woods/Mountains) reads as a bug	The asymmetric highlight + info-panel hover line	Player routes around a Mountain, or hovers 2+ terrains in one selection
Terrain defense	Same attack, different damage → suspects hidden dice	Forecast names the defender's bonus inline, every time (§2.11.3)	Resolution matches forecast, every time
The Bridge	Reads Water as decoration; "half the map is unreachable"	Briefing callout + Water never lights in any reach set + hover one-shots	First previewed or executed path crosses the Bridge
Forecast / determinism	Veterans hedge for RNG; novices fear committing	First-attack one-shot: `It is exact.`	Player commits; outcome equals card; usage becomes habitual
Positional RPS triangle	No counter chart exists — the triangle is emergent from range and move, invisible until experienced	The counter-reason line (§2.11.3), the three edge one-shots, the Artillery dead-zone ring	One uncountered Artillery strike lands — the undamaged-strike toast is the receipt
Capture	Parks a Tank on the factory and waits	Turn 2 directive; non-Infantry on a capturable tile shows disabled `Capture — Infantry only.`	Pip → tile recolors → next turn's `+100 Fame — Factory` toast
Fame income & build	Fame is abstract; player hoards, never connects factories → army	Beat 3, which holds the strip on turn 2 or turn 3 in every branch where the player has not already built — and where they have, the lesson landed without it (B); income toasts; `BUILD` pulse when affordable; greyed rows with shortfall (§2.11.5)	A bought unit spawns and stands on the board carrying its unacted pip — the **event**, not the directive, so the row also confirms for a player who skipped guidance
Flag = the game	Loses one specific Tank in a "fair trade"; match ends out of nowhere	Briefing callout; persistent `H`; red-edged info panel; flag warning band (§2.11.3)	Any lethal-to-flag forecast has been shown before any match can end
Turn cap & tiebreak	Assumes "most stuff wins"; income exclusion and passivity-draw are the least guessable rules	Scoreboard rows in tiebreak order (§2.11.4); first-combat-Fame one-shot; cap−5 banners	Chevron flips noticed in play; end screen matches the board watched all match
Repair	Heal silently fails next to an enemy; reads as a bug	Eligibility pip; the `enemy adjacent` one-shot fires at the *blocked* case first if it occurs	`+[N] HP — repaired` toast at turn start

2.11.7 Art (unchanged)

Flat/low-poly color-coded hexes, simple unit meshes or billboarded icons, generative/agent-assisted. Feel comes from subtle tweens and clear feedback, not VFX. One readability addition, cheap and worth naming: ownership is **double-coded** (faction color *and* a shape/border difference), so faction reading never depends on hue alone.

2.11.8 Build ranking (solo dev, UMG)

Must-have for a playable first session: selection state machine + input table (§2.11.1); reachable highlight and path preview (the path is already computed by §2.5 pathfinding — rendering it is cheap); forecast card with terrain-inline, counter-reason, HP before→after, kill-Fame line, and flag band; scoreboard in tiebreak order with turn counter and chevron; Fame pool + income widget; production menu with grey/shortfall/boxed states; info panel; End Turn + idle confirm; turn banner + paced AI playback with click-to-skip; toast queue; flag `H` markers and unacted pips; directive strip with the four beats; the one-shot system (string table + boolean flags); pre-match briefing overlay; end-of-match screen with tier + rows + one faction line.

Polish: chevron flip animation; Artillery dead-zone shading (plain target highlight conveys most of it); cap banners' timing tuning; camera smoothing and zoom-to-action; hotkey remap via Enhanced Input; colorblind palette variants beyond the default double-coding; hover-delay tuning; a dedicated micro-tutorial map (explicitly stretch — it would occupy a stretch scenario slot, §2.10).

2.12 Lineage extraction — what we kept, diverged, and cut from *Conflict*

The prototype was digested against the original *Conflict* (Vic Tokai, NES, 1989) manual so the lineage is a deliberate set of decisions, not an accident. **Kept and mapped:** hex board, terrain move/defense, capturable towns/factories, factory production, a single Fame currency, held-objective income, and destroy-the-commander victory. **Extracted as new pillar-fitting features:** the *Bridge* terrain (§2.3), *Repair* at owned objectives (§2.7) — the one survivable slice of *Conflict*'s logistics — *starting Fame + difficulty-as-Fame-handicap* (§2.7, §2.9), the *territorial-domination* secondary win (§2.8), and an all-1.0 *type-effectiveness* lever (§3 spec). **Deliberately diverged/cut** (recorded so the choice is on the record, not implemented): the real-time **battle mini-game** (NORMAL/AUTO + weapon/maneuver menus + evasion %) → replaced by the deterministic forecast (Pillar 1); **fuel/ammo logistics** and their supply units → cut for pacing; the **16-map password campaign** → one polished scenario; and *Conflict*'s **combat Fame penalties** (retreat/base/flag) → omitted so a flag kill can never be out-piled by attrition (Pillar 2, §1.5 #5). **Declined levers:** the per-turn **activation cap** (3-Units mode) and **one-type-per-turn** production — both lean against Pillar 2 or duplicate the existing board-space throttle.

2.13 Scenario & map design

> *All layout values are starter/tuning targets, per the GDD's standing convention (§2.3, §2.7). Nothing here adds a unit, terrain, or rule: the palette is the six movement terrains + the capturable Factory tile (§2.3) and the four-unit roster (§2.4), exactly.*

2.13.1 Layout conventions (shared by every map)

- **Coordinates:** `(col, row)`, col 0 = west, row 0 = north; pointy-top hexes (§2.2), odd rows offset +½ hex east (odd-r). ASCII glyphs: `p` Plains · `w` Woods · `m` Mountains · `~` Water · `B` Bridge · `T` Town · `F` Factory.

- **Factories:** each side owns exactly one **home factory** at start; all others are **neutral** (no income until captured, §2.7). The factory set is also the §2.8 territorial-domination win set, so factory count and placement is the primary match-length dial (see 2.13.5).

- **Starting force (standard): 5 units per side — 1 Flag Tank (§2.4, not producible), 2 Infantry, 1 Artillery, 1 Recon** — plus 200 starting Fame (§2.7) ± the §2.9 difficulty handicap. Rationale: one of each producible system is live from turn 1, so the positional RPS triangle and capture are in play before the first build; two Infantry means the standard opening (one to a town, one to a neutral factory) doesn't consume the only capturer; the producible force is worth 550 Fame — about two turns of mid-game income — so losing the opening force is recoverable, not match-ending. *No section outside §2.13 sizes the starting force; this count is pending Director approval (Q15, §4.7), not silently adopted.*

- **Home factory hex starts empty** in every deployment, so the turn-1 build spawns on the factory itself (§2.7 spawn rule) instead of scattering.

- **Validation invariants** (for the §4.2 `validate_scenario` tool — schema per §4.7 Stub 7): every land-passable hex reaches every factory (Bridges are the only Water crossings, §2.3, and there is no sea unit — connectivity is a build-time check, not a hope); all deployment hexes are free and land-passable; factory count in the map file equals the count the domination check uses; declared symmetry is machine-verified **in axial** (§4.7's T-SCN-05 forbids loaded state from holding `(col, row)` at all, so the check has no other place to run). The declarable values are **`rot180` or `none`** — `mirror` is not one of them, because no odd-r rectangle has a *vertical* mirror axis at any dimension and no map in the set is drawn to the *horizontal* one, which exists only on an odd row count (see the symmetry note at the end of this section); and `rot180` is well-formed only on an **even row count**. Both constraints are pending Q24, §4.7. Whether the horizontal mirror is worth a third enum value — it being available and merely unused rather than impossible — is **Q26**, which owns that question in full.

- **Opening-capture reachability** — the invariant the guided opening (§2.11.6-B, turn 2) rests on, and the reason it can cite a scenario guarantee at all. For **each seat**, at least one Infantry deployment hex must have a land path to a **neutral** factory costing **≤ 2 × Infantry Move = 6 movement points** (§2.4 Move 3) and crossing **no Bridge**: two turns of movement then put that Infantry on the factory hex, and the capture pip appears without making a contested crossing the first lesson. The scenario file names that unit and that factory (`guidedOpening.infantry`, `guidedOpening.objective`, §4.7 Stub 7) so the turn-1a *marked* Infantry is the one already standing on the lane. Measured on the three maps as drawn (cost in movement points, cheapest legal path, Bridge-free):

Map	West lane	East lane
Ferrum Crossing	(1,5) → South **(5,7)**, **5 MP**, all Plains	(9,3) → North **(6,2)**, **5 MP** (4 hexes, one mandatory Woods ring hex)
Longwater March	(1,2) → **(4,1)**, **4 MP** (all Plains)	(11,5) → **(8,6)**, **4 MP** (all Plains)
The Causeway	(1,2) → **(3,2)**, **3 MP** (via the town at (1,1))	(7,5) → **(5,5)**, **3 MP** (via the town at (7,6))

  All three pass. Three things the invariant deliberately does *not* promise:

  1. **"Capturing by turn 2", never "captured by turn 2."** With capture N = 1 (§2.7, Q4) the Infantry stands on the factory at the end of turn 2 and the tile **flips on turn 3**. The turn-2 directive therefore retires on the *pip*, which is exactly what §2.11.6-B already specifies.
  2. **Slack is not uniform.** *Ferrum Crossing* carries only 1 MP of slack against the 6, so a turn-1 move spent walking away from the lane pushes the pip to turn 3 — still inside the guided window, which runs **turns 1–4** (§2.11.6-B). *Longwater March* carries 2 MP of slack on both lanes, *The Causeway* 3 on both; being symmetric, each carries the *same* slack in both seats, which is what makes them usable as controlled tests. Both stretch lanes are also clear of that seat's own four other starting units, so they price identically under either reading of Q21 (§4.7) — the shipped map is the only one whose numbers a Q21 ruling could move.
  3. **Uncontested, not merely reachable.** The designated lane is the seat's *own* neutral — West → South, East → North on the shipped map — so the first lesson is not a race. Player-first IGOUGO (§2.1) also means the player occupies the hex before the AI's second turn.

Declared symmetry does imply equal lane cost — and that is what makes it worth declaring. A 180° rotation of a hex board is an isometry: it preserves hex distance, and because it maps every hex to a hex of identical terrain it preserves *path cost* too. On a genuinely symmetric map the two seats' lanes therefore cost exactly the same, and a 1 MP split is not an offset artefact — it is proof the layout is not symmetric. An earlier draft of this section argued the opposite from the odd-r row offset; that confused a storage convention with a geometry. §4.7's T-SCN-05 forbids loaded state from holding `(col, row)` at all, so the symmetry check necessarily runs in axial, where no offset exists and a declared symmetry is an isometry or it is nothing.

Two facts about offset rectangles follow, and both stretch maps are drawn to them:

1. **No odd-r rectangle has a *vertical* mirror axis, at any dimension — but one with an odd row count has a horizontal one.** Even rows occupy columns 0…W−1 and are centred on (W−1)/2; odd rows sit ½ hex east and are centred on

W/2. The centres differ by ½ hex for every W, so a single **vertical** axis cannot serve both parities, at any dimension. The horizontal axis is a different question with a different answer: c ↔ c, r ↔ H−1−r is exact whenever H−1 is even — i.e. whenever **H is odd** — because r and H−1−r then share a parity, so every row keeps its ½-hex offset and no column moves at all. In axial it is one parity-free affine map, **μ(q, r) = (q + r − (H−1)/2, H−1−r)**, integer-valued exactly when H is odd. That is the precise complement of fact 2's even-row-count precondition, so an odd-r rectangle admits *at most one* of the two symmetries and its row parity chooses which: even H → 180° rotation, odd H → horizontal mirror, never both, never a vertical mirror. 11 × 9 *Ferrum Crossing* (§2.13.2) sits on a geometry that has such an axis and is deliberately not drawn to it (§2.13.4) — an asymmetry that is a design choice, not a geometric accident. Whether a horizontal `mirror` therefore becomes a declarable value alongside `rot180`, with an odd-row-count precondition, is open as **Q26** (§4.7), which owns that question; no map in the set declares one today.

2. **An odd-r rectangle admits a 180° rotation only when the row count is even.** The rotation pairs row r with row H−1−r. With H odd those two rows share a parity, so each must land on a row carrying the same ½-hex offset — and a rotation is one rigid motion with one centre, while fact 1 has just shown the two parities' centres differ by ½ hex. Put that centre on the even rows, at (W−1)/2, and every odd row reflects as c ↔ W−2−c, which sends column W−1 to column −1. Put it on the odd rows, at W/2, and every even row reflects as c ↔ W−c, which sends column 0 to column W. Either placement throws one column of every second row off the board. In axial the same fact is pure arithmetic: the rotation constant W − H/2 is a half-integer on odd H, so *no* hex has a hex image — on 9 × 9 the constant is 4.5 and (1,1) rotates to column 6.5, which is why §4.7's T-SCN-09 refuses the file with a reason instead of reporting failed comparisons. With H even the parities alternate and the board closes exactly under **ρ(c, r) = (W−1−c, H−1−r)**. This is why *Longwater March* is 13 × **8** and *The Causeway* is 9 × **8** rather than nine rows each.

The validator still **measures and records each seat's cost as a number** (T-SCN-08) rather than reading it off the symmetry flag. Not because the flag is untrustworthy in principle, but because it is an *authored declaration* and measurement is the only thing that catches it being wrong. That is exactly what it caught here: the flag said mirrored, the numbers said 3 and 4, and the numbers were right.

2.13.2 The shipped scenario — *Ferrum Crossing* (§2.10 IN)

Spec	Value
Dimensions	**11 × 9 = 99 hexes**
Starting units per side	**5** (standard force, 2.13.1)
Factories	**4** — one home per side + 2 neutral (verbatim the §2.7 "~4 total" layout)
Towns	**4**
Turn cap	**20 turns** (the §2.8 per-scenario cap, Stub 7 `turnCap`)
Symmetry	**Asymmetric** — handicap story below
Estimated match length	**12–16 turns** (reasoning below)

Layout.

```
r0:  p  p  p  p  p  ~  p  p  p  p  p
r1:   p  p  p  T  p  B  w  p  T  p  p
r2:  p  p  w  p  p  ~  F  w  p  p  p
r3:   p  p  p  w  p  ~  w  p  p  p  p
r4:  p  F  p  p  p  B  w  w  p  F  p
r5:   p  p  w  p  p  ~  p  p  p  p  p
r6:  p  p  p  p  m  p  m  T  p  p  p
r7:   p  p  T  p  p  F  p  p  p  p  p
r8:  p  p  p  p  p  p  p  p  p  p  p
```

Key coordinates. Water (5,0)(5,2)(5,3)(5,5) · Bridges **(5,1)** north, **(5,4)** center · Home factories: West **(1,4)**, East **(9,4)** · Neutral factories: **North (6,2)** on East's bank, **South (5,7)** in the mountain pass · Towns (3,1)(8,1)(2,7)(7,6) · Pass mountains (4,6)(6,6) · Woods bands: East bank (6,1)(7,2)(6,3)(6,4)(7,4), West approaches (2,2)(3,3)(2,5).

Terrain distribution (99 hexes): Plains 75 · Woods 8 · Mountains 2 · Water 4 · Bridge 2 · Town 4 · Factory 4. Three-quarters open ground is deliberate: movement stays fast (§2.3 cost 1), contact comes early, and the match fits Pillar 2's 10–15-minute envelope — cover is a *purchase*, not the default.

Starting positions (all on Plains; home factory hex left free):

Unit	West	East
Flag Tank	(0,4)	(10,4)
Infantry ×2	(1,3), (1,5)	(9,3), (9,5)
Artillery	(0,3)	(10,5)
Recon	(0,5)	(10,3)

Terrain as economics, hex by hex. The river spans rows 0–5 only, crossed at two Bridges (−10% defense, §2.3): a unit forcing a crossing under Artillery (range 2–3, no counter, §2.4) buys ground with HP — the intended price. The river deliberately does **not** bisect the map: the southern pass (row 6) is a bridge-free route priced by two Mountains (cost 3, +40%), so it is the *slow* flank, never a free one. The one-hex river also means opposite banks are distance 2 — inside Artillery range, and the prototype ships without line-of-sight blocking (§2.2) — so bank control is contested by fire even before anyone crosses. East's bank carries the Woods band (+20%): cover in exchange for the longer road south. West gets open Plains: speed in exchange for fighting uncovered. The South neutral factory sits between the pass Mountains — +15% defense and a spawn point anchoring the southern flank; the North neutral sits on East's bank, so West must win a bridge to contest it. Rear towns (3,1)/(2,7) west and (8,1)/(7,6) east are the repair points (§2.7): the anti-fortress clause (no repair adjacent to an enemy) keeps the forward ones from healing a frontline garrison.

The tactical question this map asks: *which neutral factory do you race, and which bridge do you contest — knowing the two answers pull a 5-unit army in opposite directions?* West's fast prize is South (open Plains approach); East's is North (no crossing needed). Committing to both splits the army into halves that lose either fight.

Asymmetry and its handicap story. Starting Fame and forces are identical; the asymmetry is purely spatial and intended to be self-balancing — each seat is closer to a different neutral factory, and each seat's advantage (West: tempo and open approaches; East: cover and a crossing-free path to its prize) is the other's problem. If §4.1 self-play shows either seat above ~55% win rate, the corrective is the existing §2.9 dial — a per-seat starting-Fame offset — never a terrain rework. This is also the replay lever: the two seats are two different deterministic puzzles (see 2.13.4).

If one side holds both bridges: the other side is not locked out — the southern pass exists precisely so bridge control here is *tempo*, not a topological wall. A double-bridge holder who then sits earns zero combat Fame and loses the §2.8 cap tiebreak (or draws under the mutual-passivity guard); meanwhile the cross-river Artillery duel and the open southern pass keep combat Fame available to both sides. Full lockout as a map premise is reserved for *The Causeway* (2.13.6), where the tiebreak rules carry the whole anti-turtle load by design.

2.13.3 Match-length reasoning — and the dial it exposes

Estimate: 12–16 turns; cap at 20. Not just the number:

- **Contact turn 3–4.** Homes are 8 columns apart over mostly cost-1 Plains; Tank (move 5) reaches a bridge or the pass mouth in 2–3 turns, Infantry (move 3) in 3–4.
- **Economy online turn 3–4.** Each side's near Infantry reaches its closest neutral factory in ~2 turns; capture at N=1 (fixed from §2.7's "start N=1–2" range — Q4, §4.7) flips income the following turn. Income ramps 100 → ~225–250 Fame/turn: a reinforcing unit every 1–2 turns, continuously feeding the fight without flooding it (board space is the throttle, §2.7).
- **Kill speed.** Damage is punchy against 8–20 HP pools (§2.4): units die in 2–3 hits, so bridge and pass fights *resolve* rather than accumulate. Mid-game runs turns 5–12; the flag hunt closes 12–16.
- At ~45 s/player-turn that is ~10–13 minutes — inside §1's "10–15 typical, ~20 at the cap".

The match-length dial is factory count × home separation. More neutral factories → steeper income ramp → bloodier, faster mid-game and a wider "objectives held" spread at the cap; fewer → slower armies and more stall risk. Each column of home separation moves the contact turn back roughly one-for-Tank-move. Scenarios tune these two numbers first; unit and terrain stats are never per-map (those belong to rules and balance, §3).

2.13.4 Replayability — configurations, not mechanics

Stratocracy is deterministic end-to-end (Pillar 1, §2.6, §2.9's tier-invariant AI), so any line that beats the AI once beats it forever; by match ~4 a single mirrored map is a solved puzzle. The replay unit is therefore a **configuration = map × seat × difficulty handicap**, each a distinct deterministic puzzle:

Matches	Configuration	Ships in
1–3	*Ferrum Crossing*, West seat, Easy → Normal → Hard (§2.9 Fame ladder)	Core
4–6	*Ferrum Crossing*, **East seat**, same ladder	Core (seat-select = scenario data + one menu affordance)
7–8	*Longwater March*, Normal → Hard	Stretch P1 (§4.4 wk 4)
9–10	*The Causeway*, Normal → Hard	Stretch P2 (wk 4, only after P1)

If no stretch lands, the shipped scope alone moves the cliff from match ~3 to match ~6 — that is what the asymmetric map buys, and why *Ferrum Crossing* is not mirrored. Honest ceiling: determinism means every configuration is eventually solved; this multiplies puzzles, it does not make them unsolvable. The long-term fixes (AI second pass, map-gen MCP toolset) already sit in the GDD's stretch column and are not re-proposed.

2.13.5 Stretch scenario — *Longwater March* (P1, §4.4 wk 4)

Spec	Value
Dimensions	**13 × 8 = 104 hexes** — eight rows, not nine, because a 180° rotation only closes on an even row count (§2.13.1)
Starting units per side	5 (standard force — one variable at a time; this map's variable is factory count)
Factories	**6** — one home per side + 4 neutral *(above §2.7's "typical ~4", inside its "two or more neutral"; pending Q19, §4.7)*
Towns	4
Symmetry	**180° rotational**, ρ(c, r) = (12−c, 7−r) — fair and admittedly dull; chosen so the factory-count dial is the only variable under test. *Mirrored* was the earlier declaration and is withdrawn: no odd-r rectangle has a vertical mirror axis at any dimension, and the horizontal one exists only on an odd row count — this map is 8 rows (§2.13.1). Rotation costs this map nothing it wanted from mirroring — it is equally distance-preserving, so both seats' lanes cost 4 MP and the only asymmetry left is the one under test.
Estimated match length	**16–20 turns, frequently reaching the cap**

```
r0:  m  p  p  p  p  p  T  p  p  p  p  p  m
r1:   p  p  p  p  F  p  p  p  F  p  p  p  p
r2:  p  p  p  p  p  p  p  p  p  p  p  p  p
r3:   p  F  p  T  p  w  w  p  p  p  p  p  p
r4:  p  p  p  p  p  p  w  w  p  T  p  F  p
r5:   p  p  p  p  p  p  p  p  p  p  p  p  p
r6:  p  p  p  p  F  p  p  p  F  p  p  p  p
r7:   m  p  p  p  p  p  T  p  p  p  p  p  m
```

Key coordinates. Home factories: West **(1,3)**, East **(11,4)** — 10 hexes apart, unchanged from the 13 × 9 draft, so §2.13.3's contact-turn arithmetic is untouched. Neutral factories **(4,1)(8,1)(4,6)(8,6)** · Towns **(6,0)(3,3)(9,4)(6,7)** · central Woods knot **(5,3)(6,3)(6,4)(7,4)** · corner Mountains **(0,0)(12,0)(0,7)(12,7)**.

Every one of those is a ρ-pair under ρ(c, r) = (12−c, 7−r), and the pairing is the map's whole warrant: (1,3)↔□(11,4) homes, (4,1)↔□(8,6) and (8,1)↔□(4,6) neutrals, (3,3)↔□(9,4) and (6,0)↔□(6,7) towns, (5,3)↔□(7,4) and (6,3)↔□(6,4) woods, (0,0)↔□(12,7) and (12,0)↔□(0,7) mountains. Rows 2 and 5 are all Plains and map to each other. Note that ρ has no fixed hex on an even-row board, so the homes sit on *different rows* (3 and 4) — that is the rotation working, not a drafting slip.

Terrain distribution (104 hexes): Plains 86 · Woods 4 · Mountains 4 · Water 0 · Bridge 0

· Town 4 · Factory 6. Even more open than the shipped map (83% Plains vs. 76%), which is the point: this is the maneuver map, and the only terrain that slows anyone is the four-hex Woods knot standing between the two home rows.

Starting positions (all on Plains; home factory hex left free; East is the exact ρ-image of West, which is what "one variable at a time" means here):

Unit	West	East
Flag Tank	(0,3)	(12,4)
Infantry ×2	**(1,2)**, (1,4)	**(11,5)**, (11,3)
Artillery	(0,2)	(12,5)
Recon	(0,4)	(12,3)

Guided opening (§2.13.1): `guidedOpening.infantry` = **(1,2)** West / **(11,5)** East; `guidedOpening.objective` = **(4,1)** West / **(8,6)** East — distinct objectives, 4 MP each, no Bridge on the map at all. 2 MP of slack against the 6 MP ceiling, identical in both seats.

No Water at all — the map that teaches what the chokepoint map can't: open-field maneuver, Recon (move 7) flanking wide, and expansion tempo deciding who holds 4-of-6 **factories** at the cap. Criterion 2 counts factories *and* captured towns (§2.8), so this map's denominator is **N = 10** — 6 factories + 4 towns, against N = 8 on *Ferrum Crossing* (§2.11.4). The factory half of that spread is what the extra neutrals buy: a **0–6 factory swing inside a 10-objective sort**, where the shipped map can swing only 0–4.

Match-length reasoning: 4 neutral factories scale income toward 600+ Fame/turn (§2.7) — losses are replaced almost as fast as they land (a Tank kill pays +150 while the victim's economy rebuilds the Tank in under a turn at full income). Attrition drags, the flag hides behind rebuilt lines, and the match leans on the §2.8 tiebreak. That is the point: this is the scenario that *exercises* the combat-Fame tiebreak and the §2.11 scoreboard, which on the shipped map rarely fire.

Tactical question: *how many factories can you hold with the army those factories pay for?* Over-expansion strands Infantry on capture duty while the center Woods fight is lost; under-expansion loses the income race and, at the cap, the objectives-held sort.

2.13.6 Stretch scenario — *The Causeway* (P2, wk 4, only after P1)

Spec	Value
Dimensions	**9 × 8 = 72 hexes** — eight rows, not nine, because a 180° rotation only closes on an even row count (§2.13.1)
Starting units per side	5 (standard force)
Factories	4 — one home per side + 2 neutral (conforms to §2.7 ~4)
Towns	2
Symmetry	**180° rotational**, ρ(c, r) = (8−c, 7−r) — fair, and rotation puts each seat's near bridge on the opposite flank (West's is north at (4,2), East's is south at (4,5)), so seat-swap stays non-cosmetic even on a symmetric map. Rotation is not a preference here but the only symmetry available to an odd-r rectangle with an even row count — the vertical mirror exists at no dimension, the horizontal one only on odd H (§2.13.1); the row count is even so that the rotation actually closes.
Estimated match length	**8–12 turns**

```
r0:  p  p  p  p  ~  p  p  p  p
r1:   p  T  p  w  ~  p  p  p  p
r2:  p  p  m  F  B  p  p  p  p
r3:   p  F  p  w  ~  p  p  p  p
r4:  p  p  p  p  ~  w  p  F  p
r5:   p  p  p  p  B  F  m  p  p
r6:  p  p  p  p  ~  w  p  T  p
r7:   p  p  p  p  ~  p  p  p  p
```

Water fills column 4 end to end except Bridges **(4,2)** and **(4,5)** — **a full bisection, the deliberate opposite of Ferrum Crossing.** Column 4 is its own ρ-image, which is why the bisection survives the rotation intact. Homes **(1,3)**/**(7,4)** — 6 hexes apart, unchanged.

Neutral factories **(3,2)** and **(5,5)** each guard the approach to their adjacent bridge from their own seat's bank: +15% defense *and* a spawn point at the chokepoint (§2.7 build-and-spawn makes a held bridge-factory a reinforcement faucet exactly where reinforcements matter). Woods **(3,1)(3,3)** flank the north bridge on West's bank and **(5,4)(5,6)** flank the south bridge on East's; each seat's own crossing is the one it defends from cover, and each seat's *far* landing — (5,2) north, (3,5) south — is bare Plains, so attacking a bridge is always the uncovered half of the trade. Single Mountains **(2,2)**/**(6,5)** are Artillery perches — range 2–3 covers their bridge hex from +40% cover with no counter (§2.3, §2.4), and reaches the far landing at range 3.

ρ-pairs, exhaustively: (1,1)↔□(7,6) towns, (3,1)↔□(5,6) and (3,3)↔□(5,4) woods, (2,2)↔□(6,5) mountains, (3,2)↔□(5,5) neutral factories, (1,3)↔□(7,4) homes, (4,2)↔□(4,5) bridges, and the six Water hexes in pairs (4,0)↔□(4,7), (4,1)↔□(4,6), (4,3)↔□(4,4). Everything else is Plains.

Terrain distribution (72 hexes): Plains 52 · Woods 4 · Mountains 2 · Water 6 · Bridge 2 · Town 2 · Factory 4. The tightest board in the set, and the only one where a single terrain type — Water — decides the topology.

Starting positions (all on Plains; home factory hex left free; East is the exact ρ-image of West). *This map previously specified none, which left §2.13.1's lane figures without an antecedent and made T-SCN-06/07/08 unevaluable against it:*

Unit	West	East
Flag Tank	(0,3)	(8,4)
Infantry ×2	**(1,2)**, (1,4)	**(7,5)**, (7,3)
Artillery	(0,2)	(8,5)
Recon	(0,4)	(8,3)

Guided opening (§2.13.1): `guidedOpening.infantry` = **(1,2)** West / **(7,5)** East; `guidedOpening.objective` = **(3,2)** West / **(5,5)** East. Distinct objectives (T-SCN-07); 3 MP each, Bridge-free and confined to the seat's own bank (T-SCN-06); 3 MP of slack. The West lane is (1,2)→(1,1)→(2,1)→(3,2), through the town rather than over the Mountain at (2,2) — the 2-hex route costs 4 MP and the 3-hex route costs 3, which is the case T-SCN-08's "computes, never infers" wording exists for.

Note what the deployment does *not* do: neither seat starts within reach of a bridge. West's forward Infantry at (1,2) is 3 hexes from (4,2); the guided opening walks it away from the crossing, to a factory on its own bank. The first lesson on the bisection map is still capture, not crossing — the map earns its lockout premise from turn 3 onward, not turn 1.

Both bridges held — stated in full, because on this map it is the whole design. No naval unit, no other crossing: double-bridge control is a true lockout — the locked-out side cannot reach the enemy flag *or* home factory, so both the flag kill and territorial domination are out of its reach. The rules already make this a trap, and the map exists to prove it: the holder must still *cross* to win decisively; if it sits, it earns zero combat Fame and **loses the cap tiebreak to any opponent who dealt any damage at all** (§2.8 criterion 1); if both sides sit, the mutual-passivity guard calls a draw. Lockout on The Causeway is a tempo platform for a prepared crossing, never a victory condition — the map that demonstrates §1.5 finding #1 (the turtle exploit) is dead.

Match-length reasoning: homes only 6 columns apart, but every route crosses a Bridge; the map forces commitment, and the first successful bridgehead (turn 3–5) usually cascades into the flag kill within 4–6 turns because the defender's Fame went into defending crossings, not expanding.

Tactical question: *how do you buy a bridgehead when the crossing hex fights at −10% and the far bank fights at +20?* The §2.6 forecast makes the price exact before you pay it — the scenario where the forecast display earns its keep.

2.13.7 Scenario-set summary

Map	Status	Hexes	Units/side	Factories	Towns	Symmetry	Dial it turns
Ferrum Crossing	**Ships** (§2.10 IN)	11×9	5	4	4	Asymmetric (Fame-correctable)	baseline / seat asymmetr
Longwater March	Stretch P1 (wk 4)	13×8	5	6	4	180° rotational	factory count → cap pressure
The Causeway	Stretch P2 (wk 4)	9×8	5	4	2	180° rotational	bridge lockout – decisiven

Neither stretch map may pull work forward of week 4 or block core; if week 4 is consumed by balance (its primary §4.4 purpose), the set stays on paper.

---

3. AI Architecture — how AI agents are used (roles)

The project is run as a small **agent team directed by a human**. Each role has a clear responsibility, an instrument, and an output that is verifiable.

Role	Owner	Responsibility	Instrument	Verifiable output
Director	Human	Specs, scope line, final balance judgment, review gates, art direction, pipeline writeup	—	This GDD; merge approvals
Systems Engineer	Agent	Author the headless C++ rules (grid, movement, combat, capture, turn loop, win) from spec	Claude Code (headless module, no editor)	Compiling C++ modules
Test Engineer	Agent	Write + run the automation/unit-test suite; block merges on failures	Claude Code + test harness	Passing test suite (the merge gate)
Balance Analyst	Agent	Run headless AI-vs-AI self-play; report win-rate and turn-length; propose stat tuning	Self-play sim harness	Balance logs + tuning diffs
Content / Scenario Designer	Agent	Build maps/scenarios; generate + populate unit/terrain DataTables	Custom **MCP toolset** in-editor	Scenario assets, data tables
UI Scaffolder	Agent	Generate UMG widget skeletons + bindings for human polish	Claude Code + UE MCP	Wired UMG widgets
Opponent Commander *(stretch, runtime)*	Agent (LLM)	Play a turn in-product as the enemy	In-game LLM call + move validator	A validated legal move per turn

Pipeline. Claude Code is the agent client. Two surfaces: - **Headless harness** (Systems /

Test / Balance roles) — the rules module has **zero engine dependencies**, so these agents compile, test, and self-play in seconds without launching the editor. This is what makes agent authorship fast enough to dominate the build. - **Unreal MCP plugin** (Content / UI roles) — for in-editor operations (scenario building, UMG scaffolding).

Workflow — test-first. Director writes a spec → Test Engineer writes tests against it → Systems Engineer implements until tests pass → Balance Analyst self-plays and proposes tuning → Director reviews and approves. The agent verifies its own work against a spec, not against the Director's eyeballs.

Worked example — how a role is prompted and constrained. The role table names *what* each agent produces; the constraint is *how*. Every headless system starts as a structured **input spec** the Director hands the Systems Engineer — not a prose ask, but a contract of inputs, an exact formula, and machine-checkable invariants. For combat resolution:

```
SPEC: Combat resolution           (Director → Systems Engineer)
Inputs:  attacker{atk, type, hp, hpMax}, defender{def, type, hp},
         terrainDef% = defender's hex, attackerRange vs distance,
         eff = typeEffectiveness[attacker.type][defender.type]  (default
1.0 everywhere)
Formula: dmg = max(1, round(atk * eff * (hp/hpMax) * (1 - terrainDef%)) -
def)
Invariants (Test Engineer asserts each one):
  - terrain defense applies to the DEFENDER's hex only — never the
attacker's
  - counterattack fires ONLY if the defender survives AND the attacker is
    within the defender's range
  - Artillery (range 2–3) takes zero counter from a range-1 attacker
  - same (attacker, defender, terrain, hp) → identical result
(determinism; any RNG seeded)
Determinism: pure function of inputs, no unseeded RNG
TypeEff:     eff ∈ {0.5, 1.0, 1.5}; ships as an all-1.0 table, so RPS
stays
             positional (§2.4). Populate only if self-play shows the
triangle
             too weak. eff=1.0 everywhere → existing T-COMBAT-01..08
unaffected.
Acceptance: T-COMBAT-01..NN must pass before merge
```

The Systems Engineer is constrained to that spec; the Test Engineer turns each invariant into a gate *before* implementation exists. This is where the pipeline earns its keep — the gate catches rules the agent hallucinates. A real case:

```
T-COMBAT-07: Artillery counter-immunity
  Given  Infantry attacks Artillery from an adjacent hex (distance 1)
  Then   Artillery deals 0 counter damage        (its range is 2–3, so it
can't strike back at 1)
  First agent pass → FAIL: the Systems Engineer generalized "surviving
units counterattack"
  and wired a range-1 counter for Artillery. Merge blocked → the invariant
is re-fed →
  re-implemented → T-COMBAT-07 green → merged.
```

No human eyeballed the diff to catch that; the spec's invariant did, mechanically, at the merge gate. That is the difference between "an agent wrote some C++" and a constrained, self-verifying pipeline.

This crew is the buildable deliverable. The three headless roles above — **Systems Engineer, Test Engineer, Balance Analyst** — are exactly the 3+ coordinating agents instantiated for the standalone agent-crew build: their *game-ready output* is compiling C++ systems, a passing test suite, and self-play balance data for *this* game, handed between them in a fixed order (spec → tests → implementation → balance). The role table's I/O columns are the contract that crew is written against, so the document and the crew stay in lockstep.

Guardrails (non-negotiable). - Perforce checkpoint before any agent-driven edit session; agent permissions treated like production credentials. - The MCP plugin is **experimental, partly undocumented, and runs serially on the game thread** — it is **not on the critical path**; every editor op it does has a manual fallback. - Enable the **AllToolsets** companion plugin, or the MCP server connects but exposes nothing useful. - Claude Code is the documented client for editor-driving (not Claude Desktop).

Provenance ledger — how agent contribution is reported. The deliverable is the game (§1); agent authorship is reported honestly, not as a single hero percentage but as a **per-system ledger** — one row per system, marked by author and verification status, **each Verified row citing the commit and passing test IDs that back it** so the claim is auditable rather than asserted.

Status: live tracker — first rows populated 2026-07-26 from the Assignment-3 agent crew; ***Repair*** *and* ***Type-effectiveness*** *added and gate-verified 2026-07-29; the rest land as each system is built (wk 1–3, §4.4).* Legend: **Author** ∈ {agent, agent+human, human}; **Agent-verified** ∈ {✓, —}; **Evidence** cites the git commit + passing test IDs so any row is independently checkable.

System	Author	Agent-verified?	Evidence (commit · tests)
Combat resolution	agent	✓	`Combat.cpp` + `test_combat.cpp` @ `5ffa8d6` · T-COMBAT-01..10 (10/10)
Test suite	agent	✓	`test_combat.cpp` @ `5ffa8d6` · 17/17 invariants, re-runnable via `python run.py`
Hex grid & math	*pending*	—	*pending build*
Movement & pathfinding	*pending*	—	*pending build*
Capture & Fame economy	*pending*	—	*pending build*
Turn loop & win / tiebreak	*pending*	—	*pending build*
Opponent AI	*pending*	—	*pending build*
Data tables (units/terrain)	*pending*	—	*pending build*
Repair (owned-tile heal, §2.7)	agent	✓	`Combat.cpp::repairAmount` @ `5ffa8d6` · T-REPAIR-01..07 (7/7)
Type-effectiveness (§3 spec)	agent	✓	`Combat.cpp::effectiveness` @ `5ffa8d6` · T-COMBAT-09..10 (neutral, 2/2)
Content / scenario	*pending*	—	*pending build*
UI	*pending*	—	*pending build*

Four rows are now populated — the headless Combat module built for the Assignment-3 agent crew (github.com/jakemartin/stratocracy-crew, commit `5ffa8d6`): **Combat resolution** and its **Test suite**, plus **Repair** and **Type-effectiveness**, all agent-authored from `spec/combat_spec.md` (+ `combat_spec_addendum.md`) and verified by the real compile+test gate — a live CrewAI run authored the module and the Test Engineer certified **17/17** on a live `g++`/`clang++` compile+run. *(Commit `5ffa8d6` is published: it is an ancestor of `main` at `2fcbf32` on the public remote, so every link above resolves and the ledger's "independently checkable" claim is testable by clicking it.)* The remaining rows fill in the same format — commit + passing test IDs — as each system clears the gate.

Rules that keep the ledger honest: - **Scope = the game's own source only.** The UE engine, the MCP/AllToolsets plugins, and the bought reference template (§4.3, not shipped) are excluded — they aren't the artifact we built, so they're not in the denominator. - **Author = human** for anything a human hand-wrote or substantially edited (UI polish, glue). Human work counts as human, so the ledger under-claims rather than inflates. - **Agent-verified = both bars:** (a) covered by agent-authored automated tests that pass in the headless suite (§4.1) **and** (b) signed off at the human review gate (§3). A system is "agent-verified" only if it clears both.

Because the game is deterministic rules + data, this ledger is *checkable*: the test suite is re-runnable and each system's provenance traces to its commits. It is supporting evidence of the method — the game itself is what ships and is judged.

---

4. Technical Strategy

4.1 Architecture

- **Headless rules module (C++)** — all of §2 mechanics, no engine deps. Enables sub-second agent test/tune loops and deterministic self-play. Combat damage is a deterministic function of attacker/defender stats, terrain defense, and the attacker's HP ratio; reachability and routing use Dijkstra/A* over variable terrain cost; any RNG is seeded so tests and self-play stay reproducible.
- **Presentation layer** — Actors + UMG read from the headless state; never own rules.

- **Data-driven** — `DataTable`/`DataAsset` for units and terrain; agent-generated, human-reviewed.
- **Test suite** — Unreal Automation tests (or a plain C++ harness on the headless module) covering hex math, pathfinding, combat, capture, and win detection.
- **Balance sim harness** — headless AI-vs-AI self-play, N games → win-rate and turn-length distributions → tuning input.
- **Input:** Enhanced Input. **Save/load:** minimal single-slot for the prototype.

4.2 Engine & tooling

- **Unreal Engine 5.8** (last planned UE5 release; stable base).
- **Unreal MCP plugin** (ships in 5.8, Experimental) + **AllToolsets** — editor operations for the Content/UI agent roles. Serial-on-game-thread; off the critical path.
- **Custom MCP toolset** — `place_terrain`, `place_unit`, `set_objective`, `validate_scenario`, authored via `ToolsetDefinition` (Python or C++) so the Content agent builds/edits scenarios in-editor.
- **Reference implementation** — cross-check the agent's grid/pathfinding against a mature template (e.g. *Advanced Turn Based Tile Toolkit*). **License caveat: verify AI-usage terms before an agent reads template code.**

4.3 Build approach

Author the **core in C++ with the agent** (the on-thesis path — hex math, pathfinding, and combat are where agents are strongest and least likely to hallucinate). Keep a bought template installed **only as a reference to check output against**, not as the shipped codebase, so the "agents built the game" claim holds.

4.4 Milestones (7 weeks)

Wk	Goal
1	Headless C++ core (grid, units, move, combat, win) **+ test suite.** Playable via debug commands. **UI-scaffolder agent starts UMG widget skeletons in parallel** — the whole game is played through UI, so it can't wait until wk 2.
2	Engine presentation + UI wiring (select/move/attack) onto the wk-1 skeletons, plus the one scenario loading, validating and rendering. **Move + attack only** — no capture, no production, no AI opponent.
3	Capture + production (§4.11 rows 4–5), then the baseline objective-seeker AI (row 6) → **working vertical slice.** AI second pass (utility + threat map) and the custom MCP scenario toolset follow only if the slice lands early; both are off the critical path (§4.2, §4.11).
4	Self-play balance sims and tuning; scenario polish (additional scenarios only as stretch).
5	UI polish, feedback/juice, save/load, onboarding.
6	Playtest, bug fix, balance lock. LLM-commander stretch **only if on track.**
7	Final polish, **package/build the shippable game** (the deliverable), and write up the agent pipeline + provenance ledger (supporting evidence).

On weeks 2–3 (Q23, ruled). The vertical slice sits in week 3, not week 2, because §4.11's critical path is `1 → 3 → 4 → 5 → 6/8`: the baseline AI (row 6) needs the turn loop (row 5), which needs Capture & Fame (row 4). A week-2 slice *with* an AI opponent would have required building rows 4–6 a week before §2.10 says capture and production land — the document previously asserted both and could not have delivered either. Week 2 therefore delivers move + attack against a static board, which is the honest subset of the slice that rows 1–3 actually support, and §2.10's *"these land wk 3, not wk 1–2"* now describes the schedule rather than contradicting it. The save/replay half of the same question (**Q20**) is still open.

4.5 Risks & mitigations

Risk	Mitigation
Scope creep (16 scenarios is a commercial game)	Hard MVP line; 1 scenario + flag win is complete
UI underestimated	Treat as core; start week 2
AI rabbit hole	Ship the heuristic; LLM behind a toggle
MCP plugin experimental/flaky	Off the critical path; Perforce checkpoints; manual fallback
Pacing (the original's flaw)	Small maps, decisive damage, turn cap + tiebreak
Agent code quality	Test-first; human review gates; headless module keeps loops fast

4.6 Token budget

AI-agent work is the project's main variable cost, so it is budgeted explicitly. Two surfaces consume tokens: **development-time authoring** (agents writing, testing, and tuning the game across the 7 weeks) and the **runtime LLM commander** (a stretch feature, billed only if it ships). Rates below are Anthropic's official published API prices, **verified against platform.claude.com on 23 Jul 2026** — Sonnet 5 introductory ($2/$10 per M in/out, $0.20/M cache read), Opus 4.8 ($5/$25, $0.50/M cache read), Haiku 4.5 ($1/$5, $0.10/M cache read). *Note: the Sonnet 5 introductory rate holds **through 31 Aug 2026**, then reverts to the $3/$15 standard rate; the 7-week jam (Jul–late Aug 2026) falls entirely inside the introductory window, so these figures apply for the whole project.* As a conservative buffer, estimates carry a **~30% token-overhead margin** on top of raw text length (accounting for tokenizer overhead, tool-call framing, and re-reads). **Prompt caching** (cache reads billed at 0.1× the input rate) is assumed for the stable spec/rules context the agents re-read on every task.

A. Development-time authoring — the bulk. Workhorse model **Claude Sonnet 5** ($2 / $10 per M input/output, $0.20/M cache read), with occasional **Opus 4.8** escalation ($5 / $25 per M) for hard reasoning. The unit of work is one substantial agent task — author or iterate a system together with its tests — spanning many tool calls and file reads.

Line item	Estimate
Tokens per substantial task	~300k input (≈⅔ from cache: ~100k fresh + ~200k cache read) + ~45k output = **345k**. The ~30% overhead margin above is already inside these figures and is not applied again below
Cost per task — Sonnet 5	100k × \$2/M + 200k × \$0.20/M + 45k × \$10/M = \$0.20 + \$0.04 + \$0.45 = **\$0.69**
Cost per task — Opus 4.8	100k × \$5/M + 200k × \$0.50/M + 45k × \$25/M = \$0.50 + \$0.10 + \$1.125 = **\$1.725** (quoted to the cent as \$1.73; **\$1.725 is the exact figure**, and every escalation delta in this section is computed from it, not from the rounded display)
Task volume	5 agent roles × ~6 tasks/wk × 7 wk ≈ **210 tasks**
Sonnet-only base	210 × \$0.69 ≈ **\$145** · 210 × 345k ≈ **72M tokens**
Opus escalation — a line, not a multiplier	~15% of tasks — 15% of 210 = 31.5, i.e. **32** to the nearest whole task — run on Opus *instead of* Sonnet, same tokens at a higher rate. The delta is taken **unrounded**: \$1.725 − \$0.690 = **\$1.035/task**. So 32 × \$1.035 = \$33.12 ≈ **+\$33**. \$1.035 is the only escalation delta used anywhere in §4.6
Dev-time subtotal	**≈ 72M tokens · ≈ \$178** — unrounded, \$144.90 + \$33.12 = **\$178.02**; rounding first (\$145 + \$33) reaches the same \$178, so the printed figure is stable either way. Downstream lines scale the **unrounded** per-task costs, never this rounded total

The escalation is priced once. It is a *delta* between two models on the same task, so it belongs inside the subtotal and there is no further uplift after it — an earlier draft folded it into the subtotal silently and then applied it a second time to reach \$225, which is why three figures in one table disagreed. Every number above is re-derivable from the two rate lines and the task count, which is the property that made the error visible. The escalation assumes Opus **substitutes** for Sonnet on those tasks; if it instead runs *in addition* (a re-run rather than a substitution) the line is 32 × \$1.725 = \$55.20 ≈ **+\$55** and the subtotal \$144.90 + \$55.20 = \$200.10 ≈ **\$200**.

B. Runtime LLM commander — stretch, per match. Model **Claude Haiku 4.5** (\$1 / \$5 per M) for turn latency; Sonnet 5 as a quality upgrade. Each AI turn serialises the board and its legal moves, asks the model to rank one, then validates and applies it (§2.9).

Line item	Estimate
Tokens per AI turn	~2.5k fresh input + ~1.5k cached rules + ~0.4k output = **4.4k**
Cost per AI turn	2.5k × \$1/M + 1.5k × \$0.10/M + 0.4k × \$5/M ≈ **\$0.0047** (Haiku 4.5)
Per match (~20 AI turns)	88k tokens · ≈ **\$0.09**
200 self-play + playtest matches	≈ **17.6M tokens** · ≈ **\$19** (Haiku 4.5)
Same volume on Sonnet 5	≈ **\$37** at the introductory \$2/\$10 rate this project runs inside · ≈ **\$56** at the \$3/\$15 standard rate that resumes 1 Sep 2026

Headline. Development authoring dominates. Dev-time alone is **≈ 72M tokens · ≈ \$178**; the stretch runtime commander adds **≈ 18M tokens** and **\$19 (Haiku 4.5)** to **\$37 (Sonnet 5)**. The whole jam therefore lands at **≈ 90M tokens** and **≈ \$178 without the commander, ≈ \$197–\$215 with it**. Cost scales linearly with task volume, and the overrun case is stated rather than buried in a wide band. At **1.5× task volume** — 315 tasks rather than 210, of which 15% escalate to Opus (47.25, i.e. **47** to the nearest whole task) — the dev-time line is 315 × \$0.69 + 47 × \$1.035 = \$217.35 + \$48.645 = **\$265.995**, i.e. **≈ \$266**. The overrun is in *agent task volume*; it does not scale the 200-match playtest plan, so part B carries across unchanged and the all-in overrun is **\$266 + \$19 ≈ \$285** with the Haiku 4.5 commander and **\$266 + \$37 ≈ \$303** with Sonnet 5. **\$303 is the ceiling.** Each of those is a derivation rather than a round number — as is every figure in the two tables above. (An earlier draft printed **\$267** here. That was 1.5 × the *rounded* \$178 subtotal, not a re-derivation from the per-task lines, and it sat $1.005 above what its own stated inputs produced ($267 − $265.995; the often-quoted $1.24 came from the retired $1.03 delta) — the second arithmetic fault this table has surfaced by being fully re-derivable, and the reason the escalation delta is now quoted unrounded at $1.035 throughout.)

4.7 Pending-system gate plan — the eight `*pending*` ledger rows

Each stub below follows the proven Combat-spec shape (§3): Inputs, Formula or state transition, Invariants (one per assertable rule), Determinism, Acceptance test IDs. All rules code is **headless** — `namespace strat`, pure C++17, zero engine dependencies, compiled by the same `g++/clang++` + `python run.py` gate that certified Combat (§3 ledger). Where a stub needs a rule the GDD does not state, the gate is parameterized on a numbered open question (Q1–Q27, Open questions below) — the Director rules, the gate then pins the ruling.

Shared conventions (Director-owned contract):

- **Coordinates:** axial `(q, r)`, pointy-top hexes (§2.2). Distance is the standard axial hex metric `(|dq| + |dr| + |dq + dr|) / 2` — pure integer math. **Two conventions coexist deliberately, and the conversion between them is part of this contract:** §2.13's authored maps use odd-r offset `(col, row)`, which is what a human reads off an ASCII grid, while the module uses axial because the distance metric above is only clean in axial. The scenario loader (Stub 7) converts odd-r → axial on import — `q = col - (row - (row & 1)) / 2`, `r = row` — and that conversion is itself gated (T-SCN-05). Neither convention leaks into the other's layer: no authored file stores axial, and no module code stores `(col, row)`.
- **Canonical hex order** (the subject of T-HEX-07): a total order over hexes — **ascending `r`, then ascending `q`**. Used everywhere enumeration order could leak into behavior or bytes: reachable-set enumeration, movement/AI tie-breaks (T-MOVE-04, T-AI-06), scenario serialization and hashing (Stub 7), and the §4.10 canonical state hash.
- **Numbering:** stubs 1–8 below are also the build-order row numbers (Build order table).

```
SPEC STUB 1: Hex grid & math               (Director → Systems Engineer)
Inputs:  map dimensions (Q1); axial coords (q, r).
Functions: neighbors(hex) — the six adjacent hexes in a fixed, documented
         enumeration order, out-of-bounds candidates filtered; distance(a,
b);
         inBounds(hex).
Invariants:
  T-HEX-01  every in-bounds hex has exactly six neighbor candidates in the
fixed
            order; filtering removes only out-of-bounds hexes (§2.2 "six
equal
            neighbours")
  T-HEX-02  distance is a metric: d(a,a)=0; d(a,b)=d(b,a); triangle
inequality
  T-HEX-03  d(a,b)=1  ▯  b ∈ neighbors(a)
  T-HEX-04  direction fairness: each of the six unit steps has distance
exactly
            1 — no direction is cheap the way diagonals are on squares
(§2.2)
  T-HEX-05  inBounds agrees with the Q1 dimensions; every hex reference
the
            engine or a scenario hands the module is bounds-checked, never
trusted
  T-HEX-06  single distance definition: combat's range checks (T-COMBAT-
06..08 @
            5ffa8d6) consume THIS distance function — Artillery at
distance 1
            cannot counter, per the verified module, with no second metric
to drift
  T-HEX-07  canonical order + determinism: sorting any hex set by (r asc,
q asc)
            is total, stable, and platform-independent; neighbors()
enumeration
            order is fixed across runs and compilers
Determinism: pure integer functions; no state.
Acceptance: T-HEX-01..07.
```

```
SPEC STUB 2: Data tables (units/terrain)   (Director → Systems Engineer)
Defined in full in §4.8 (schemas, both-side structs, parity gate).
Invariants
T-DATA-01..06 are specified there once — this stub defers to §4.8 rather
than
duplicate a contract that must never fork.
```

```
SPEC STUB 3: Movement & pathfinding        (Director → Systems Engineer)
Inputs:  unit {move, moveClass}; terrain move costs (§2.3 via Stub 2);
current
         occupancy; start hex.
Transition: reachable set + cheapest paths via Dijkstra over terrain cost
(§4.1);
         executing a move relocates the unit along the chosen path (§2.5).
Invariants:
  T-MOVE-01  reachable set is exact: a hex is in the set ⟺ its cheapest
path
             cost ≤ Move — "the real move set, not an estimate" (§2.5)
  T-MOVE-02  costs per §2.3: Plains 1, Woods 2, Mountains 3,
Town/Bridge/Factory
             1; Water impassable to land — a land path across a Water span
             exists ⟺ it crosses on Bridge hexes (§2.3, "the only hex a
land
             unit crosses Water")
  T-MOVE-03  occupancy: a move never ends on an occupied hex (§2.5, one
unit per
             hex). Pass-through of friendly-occupied hexes: parameterized
on the
             Q3 ruling; until ruled, the gate asserts the conservative
reading
             (occupied hexes block pathing entirely)
  T-MOVE-04  the executed path is minimal-cost; ties between equal-cost
paths are
             broken by canonical hex order, so the route is reproducible
  T-MOVE-05  no zones of control: moving adjacent to an enemy costs
nothing
             extra and freezes nothing (§2.5 — ZOC is cut)
  T-MOVE-06  determinism: same state → identical reachable set and
identical path
  (T-MOVE-07 reserved: Recon's "ignores some terrain cost" (§2.4) —
blocked on
   the Q2 movement-class ruling; no gate is written until the rule
exists.)
Determinism: pure; all tie-breaks canonical.
Acceptance: T-MOVE-01..06 (07 reserved on Q2).

SPEC STUB 4: Capture & Fame economy        (Director → Systems Engineer)
Inputs:  game state; commands Build{factoryHex, unitId}, Capture{unit};
the §2.7
         income/award values; §2.4 costs via Stub 2.
Transition: income accrual; build-and-spawn; capture progress; kill
awards.
Invariants:
  T-FAME-01  single pool (§2.7): income, kill awards, and spending all
mutate one
             per-side fameTotal; combat awards ALSO accrue a separate
fameCombat
             counter (the §2.8 tiebreak criterion-1 sort key); passive
income
             never touches fameCombat
  T-FAME-02  income: each held factory pays +100/turn, each held town
+25/turn
             (§2.7); accrual timing per the Q8 ruling
  T-FAME-03  build: deducts the exact §2.4 cost (Infantry 100, Recon 150,
             Artillery 200, Tank 300); refused if unaffordable; fameTotal
is
             never negative
  T-FAME-04  spawn: on the factory hex if free, else an adjacent free hex,
else
             the build waits (§2.7); waiting-build semantics per Q8
  T-FAME-05  capture: Infantry only (§2.7, §2.4); completes after N turns
of
             holding (N per Q4); interruption/reset semantics per Q4
  T-FAME-06  a captured objective's income flips to the new owner (§2.7)
  T-FAME-07  kill awards: ~half the victim's Fame cost, small undamaged-
strike
             bonus, flag kill +500 and the match ends (§2.7) — exact
values per
             Q5/Q6; the gate pins whatever the Director rules
  T-FAME-08  no Fame cap: fameTotal is unbounded; deployment is throttled
by
             board space only (§2.7)
  T-FAME-09  determinism: same state + command → identical Fame deltas and
state
Determinism: pure state transitions; no RNG anywhere in the economy.
Acceptance: T-FAME-01..09.
```

```
SPEC STUB 5: Turn loop & win / tiebreak    (Director → Systems Engineer)
Inputs:  game state; per-unit act flags; the turn counter and cap (Q7,
stored in
         the scenario file, Stub 7); commands incl. EndTurn{}.
Transition: I-GO-U-GO alternation (§2.1); win/loss/draw evaluation (§2.8);
         start-of-turn repair application (§2.7).
Invariants:
  T-TURN-01  strict alternation; each unit acts at most once per own turn,
in
             any order the owner chooses (§2.1)
  T-TURN-02  flag death ends the match immediately — Decisive win for the
killer,
             loss for the owner (§2.8)
  T-TURN-03  territorial domination: controlling every factory on the map
at the
             start of your turn ends the match immediately, ranked
Decisive
             (§2.8; factories only, towns excluded)
  T-TURN-04  at the turn cap, the attrition tiebreak resolves in the exact
§2.8
             order: combat Fame → objectives held → surviving HP → draw
  T-TURN-05  mutual-passivity guard: both sides' fameCombat == 0 at the
cap →
             immediate draw, with NO fall-through to objectives held
(§2.8)
  T-TURN-06  criterion 2 (objectives held, X of N) is evaluated only when
both
             sides fought and their fameCombat is equal (§2.8)
  T-TURN-07  result tiers are categorical: Decisive > Marginal > Draw,
regardless
             of Fame totals — Fame is only the sort key inside criterion 1
(§2.8)
  T-TURN-08  repair fires at the start of the unit's turn exactly when the
             verified repairAmount says so (owned Town/Factory, no
adjacent
             enemy, +25% max HP floored, min 1, capped — T-REPAIR-01..07 @
             5ffa8d6); this gate asserts the turn loop calls it at the
right
             moment with the right board facts, nothing more
  T-TURN-09  determinism: the same command sequence from the same scenario
→
             identical result tier and identical state at every step
Determinism: pure state machine; the §4.10 hash is taken from this state.
Acceptance: T-TURN-01..09.

SPEC STUB 6: Opponent AI (baseline)        (Director → Systems Engineer)
Inputs:  full game state (the AI cheats at nothing and sees only real
state);
         the §2.9 baseline routine; default buildlist (§2.9).
Transition: economy phase then unit phase, emitting ordinary commands that
the
         rules module validates like any player's (§2.9).
Invariants:
  T-AI-01  legality: every AI command passes the same validation as a
player
           command; zero rejected commands across N self-play games
  T-AI-02  economy phase: at each held factory, if a buildlist unit is
           affordable, one is built — the AI spends and replaces losses
rather
           than hoarding (§2.9)
  T-AI-03  capture behavior: an idle Infantry near an uncaptured,
undefended
           factory/town moves onto it to capture (§2.9) — objectives stay
live
           on both sides
  T-AI-04  attack preference: the enemy flag if in reach, else the best
           expected-damage target; Artillery fires from maximum standoff
(§2.9)
  T-AI-05  strictly-losing-attack guard: the AI never makes an attack in
which
           its unit dies and trades down (§2.9)
  T-AI-06  determinism: same state → same move; every scoring tie is
broken by a
           stated deterministic rule (canonical hex order for position
ties;
           remaining tie dimensions per the Q9 ruling)
Determinism: pure function of state; difficulty changes only starting Fame
         (§2.9), never the routine.
Acceptance: T-AI-01..06, plus a self-play smoke run: N headless AI-vs-AI
games
         all terminate at or before the cap with a valid result tier.

SPEC STUB 7: Scenario file & validator     (Director → Content/Scenario
Designer,
                                            Systems Engineer for the
```

```
                                          Systems Engineer for the
loader)
Format:  one versioned JSON file per scenario; strat::loadScenario parses
and
         validates it headless. Fields:
           formatVersion   int     unknown → refuse load
           scenarioId      string  stable identifier
           scenarioHash    string  hash of the canonical serialization
(fields
                                   in this order, hexes in canonical hex
order).
                                   The order is load-bearing, so NEW
FIELDS
                                   APPEND AT THE TAIL of this list: an
                                   append-only preimage means adding a
field can
                                   never reorder an existing one (the
discipline
                                   that appended `type` last in the combat
                                   addendum, Part A). Serialization order
is not
                                   validation order — a field is validated
after
                                   whatever it references, wherever it
sits here.
           map             object  width/height (Q1) + per-hex terrain Id
                                   (row-major in canonical hex order; Ids
are
                                   §2.3 / Stub 2 row names)
           ownership       array   initial owner of each capturable hex —
a home
                                   factory per side, neutral
factories/towns
                                   unowned (§2.7)
           placements      array   {side, unitId, hex, isFlag} — starting
units;
                                   isFlag is valid only on a Tank (§2.4:
the
                                   flag is "a designated Tank")
           startingFame    object  per side; 200/200 baseline (§2.7). The
                                   difficulty handicap (§2.9) is a match-
setup
                                   parameter applied on top, not a
scenario field
           turnCap         int     the §2.8 cap (value per Q7)
           guidedOpening   array   one entry per side, {side, infantry,
                                   objective} — that seat's opening-
capture lane
                                   (§2.13.1). `infantry` and `objective`
are HEX
                                   references: the deployment hex of the
seat's
                                   marked Infantry (§2.11.6 turn 1a) and
the
                                   neutral Factory hex it walks to. A hex
                                   identifies the placement uniquely
because
                                   T-SCN-02 already forbids two placements
                                   sharing one, so this adds no instance-
ID
                                   concept to the schema. Entries
serialize in
                                   the module's side enumeration order,
not
                                   authoring order, so the hash is
content-only.
                                   Required on every scenario: §2.13.1
declares
                                   the reachability invariant shared by
every
                                   map, and §2.11.6's turn-2 directive has
no
                                   other source for the pair. The guidance
layer
                                   reads this field from the loaded
scenario
                                   directly — marked/locked is
presentation
                                   state, not rules state, so it stays out
of
                                   the Stub-8 snapshot.
           symmetry        enum    REQUIRED. `rot180` or `none`, the two
values
                                   §2.13.1 declares (pending Q24). The
enum is
```

```
                                   narrow by SCOPE, not by impossibility,
and
                                   the difference is load-bearing. An odd-
r
                                   rectangle has no VERTICAL mirror axis
at any
                                   dimension (§2.13.1) — but on an ODD row
                                   count it does have a HORIZONTAL one,
                                   mu(c, r) = (c, H-1-r): with H odd, H-1
is
                                   even, so r and H-1-r share a parity,
every
                                   row keeps its offset and the column is
                                   unchanged. 11 x 9 Ferrum Crossing
carries
                                   that axis geometrically and still
declares
                                   `none`, because its terrain is not
drawn to
                                   it. So `mirror` is a claim the schema
gives
                                   an author no way to STATE, not one the
                                   validator could never ACCEPT — two
different
                                   failure modes, and Q26 asks which one
is
                                   intended. Admitting the value could
widen
                                   nothing: rot180 needs an even H and a
                                   horizontal mirror an odd one, their
                                   composition would be the vertical
mirror
                                   that never exists, so at most ONE non-
`none`
                                   value is well-formed on any given map.
                                   Absent or unrecognized is a hard load
                                   failure, never a default of `none` — a
                                   scenario that forgets to declare must
not
                                   silently claim the weakest claim. This
is
                                   not a balance or layout field: it is
the
                                   AUTHORED CLAIM that T-SCN-09 verifies
hex by
                                   hex, and it exists in the schema
because
                                   §2.13.1 asks validate_scenario to
                                   machine-verify a declaration the file
                                   previously had no place to make.
Appended at
                                   the tail per the scenarioHash policy
above;
                                   it moves every scenario's hash, which
costs
                                   nothing while no scenario file exists
and is
                                   the point of the policy once one does.
Invariants:
  T-SCN-01  exactly one isFlag placement per side, and it is a Tank; the
flag
            appears in no buildlist anywhere — "not producible" (§2.4) is
a
            scenario/production fact, not a fifth unit row
  T-SCN-02  structural validity: every hex reference is in bounds (T-HEX-
05);
            every terrain and unit Id resolves to a Stub-2 row; no two
            placements share a hex (§2.5)
  T-SCN-03  economy validity: each side owns exactly one home factory at
start,
            and at least two neutral factories exist in contested ground
(§2.7,
            "~4 factories total")
  T-SCN-04  playability: the two flags are mutually reachable by land
movement
            (Stub 3 pathing, Bridge rules respected) — a scenario cannot
be born
            stalemated
  T-SCN-05  coordinate conversion (Shared conventions): odd-r (col, row) →
axial
            (q, r) round-trips for every in-bounds hex of the declared
            dimensions, and the loaded map's adjacency matches the
authored
            grid's: no authored file stores axial, and no loaded state
```

```
              grid 3, no authored file stores axial, and no loaded state
stores
              (col, row)
  T-SCN-06  opening-capture lane (§2.13.1): for EACH guidedOpening entry,
a land
              path exists from `infantry` to `objective` costing
              <= 2 x Move of the capturing unit row — T-DATA-03's single
              CanCapture row, §2.4 Infantry Move 3, so 6 MP — and crossing
NO
              Bridge hex. The ceiling is DERIVED from the loaded table,
never a
              literal: a §2.4 Move change re-prices the gate instead of
silently
              passing it, and the capturing row is found by CanCapture so
§2.7's
              Infantry-only rule and this lane can never name different
units.
              Cost counts every hex entered including the objective itself
              (Factory MoveCost 1, §2.3) — the same accounting as T-MOVE-01,
so
              the validator and the reach highlight cannot price one lane
two
              ways. Excluding Bridge (and Water being land-impassable, §2.3)
              confines the lane to the seat's own bank, which is what makes
the
              first lesson uncontested rather than a crossing. Asserting the
              existential over the NAMED hex is deliberate. §2.13.1 states
an
              EXISTENTIAL ("at least one Infantry deployment hex must have a
              land path...") and, alongside it, a NAMING requirement (the
              scenario file names that unit, so §2.11.6's turn-1a marked
              Infantry is the one already standing on the lane); quantifying
              over the named hex collapses both into a single check. Written
              the other way round — find ANY qualifying hex, then compare it
to
              the named one — the validator passes a map whose qualifying
lane
              belongs to a unit turn 1a never marks: the reachability rule
              satisfied to the letter while the marked Infantry stands
              somewhere else, and the guided opening's first directive
walking
              the player down a lane nobody priced.
              The gate asserts ARRIVAL ONLY: the turn the tile flips is
              N-dependent (Q4) and is asserted nowhere here, matching
§2.13.1's
              "capturing by turn 2, never captured by turn 2."
  T-SCN-07  opening-capture naming (structural; no pathing, so this half
of the
              check lands with rows 1-2): exactly one guidedOpening entry
per side;
              `infantry` is the hex of a starting placement of THAT side
whose
              unitId is the CanCapture row and whose isFlag is false;
`objective`
              is a Factory hex that `ownership` leaves neutral (§2.7); and
the two
              entries name DIFFERENT objectives — §2.13.1's "the seat's own
              neutral," gated at its distinctness floor. The stronger
              non-contention property §2.13.1 also claims is unruled (Q22)
and
              nothing here asserts it.
  T-SCN-08  measured, not inferred: the validator COMPUTES and REPORTS
each
              entry's lane cost as an integer, from Stub-3 pathing. The
declared-
              symmetry flag (§2.13.1) is not an input and cannot substitute.
Not
              because the flag is unreliable in principle — a verified
rot180 IS
              an isometry and does imply equal cost between IMAGE lanes
(§2.13.1,
              T-SCN-09) — but for three reasons that hold even when it is
right:
              (i) it is an AUTHORED DECLARATION, and measurement is the only
              thing that catches the author declaring the wrong one, which
is
              what happened here (the flag said mirrored; the numbers said 3
and
              4; the numbers were right, §2.13.1); (ii) a `none`-declared
map
              offers nothing to infer from at all, and `none` is what the
SHIPPED
              map declares (§2.13.2); (iii) even a verified rot180 forces
equal
              lane cost only if the two guidedOpening entries are themselves
```

```
            rho-images, which §2.13.1 does not require and T-SCN-09 does
not
            assert (Q25) — so equal numbers on a symmetric map are a
result,
            not a theorem. And no symmetry argument of any kind prices a
lane,
            because cheapest path is not fewest hexes.
            Fixtures:
              (a) The Causeway (§2.13.6) PASSES, reporting 3 and 3. Its
West
                  lane (1,2)->(1,1)->(2,1)->(3,2) is THREE hexes costing 3
MP
                  through the town; the TWO-hex route over the Mountain at
                  (2,2) costs 4 (§2.3). An implementation that counts
hexes
                  instead of summing MoveCost reports 2, and nothing else
in
                  this suite catches it.
              (b) Longwater March (§2.13.5), rot180 on 13 x 8, PASSES and
                  reports 4 and 4 — COMPUTED equal, never assumed equal.
              (c) A scenario whose lanes both cost 7 FAILS the T-SCN-06
                  ceiling, and the refusal reason CARRIES BOTH MEASURED
                  INTEGERS: an author needs to read 7 against 6, not
merely
                  "too far" (Determinism, "refuses with a reason").
            The reported integers are the source of truth for §2.13.1's
lane
            table, so a map edit that lengthens a lane surfaces as a
changed
            number rather than a still-green boolean.
  T-SCN-09  declared symmetry is VERIFIED, not trusted — the gate
§2.13.1's
            "machine-verified in axial" clause names, and which no other
            invariant asserts (T-SCN-05 supplies the axial frame and
asserts
            nothing about symmetry).
            `symmetry` == `none` asserts nothing and is always well-
formed.
            `symmetry` == `rot180` asserts, for EVERY in-bounds hex h:
              - terrain(rho(h)) == terrain(h);
              - ownership maps onto itself with the two sides EXCHANGED —
each
                home factory's image is the other side's home, each
neutral
                capturable's image is neutral (§2.7);
              - the placement set maps onto itself with sides exchanged
and
                unitId and isFlag preserved, which is what §2.13.5 and
§2.13.6
                mean by "East is the exact rho-image of West."
            SCOPE per Q25: guidedOpening is NOT bound, so a rot180 map may
name
            lanes that are not each other's image and report unequal costs
            (T-SCN-08 (iii)). Both stretch maps satisfy the stronger
reading as
            drawn, so ruling Q25 either way moves no layout.
            rho RUNS IN AXIAL, after the T-SCN-05 conversion, because no
loaded
            state holds (col, row) to rotate. On an even row count §2.13's
            authored rho(c, r) = (W-1-c, H-1-r) and the 180-degree
isometry are
            the same map, and in axial it is one parity-free affine map:
                rho(q, r) = (W - H/2 - q,  H - 1 - r)
            The EVEN ROW COUNT is a PRECONDITION, not a comparison: on odd
H
            that constant W - H/2 is a half-integer, so no hex has a hex
image
            and the file is REFUSED WITH A REASON before any comparison
runs
            (§2.13.1, pending Q24). On 9 x 9 the constant is 4.5 and (1,1)
            rotates to column 6.5 — one refusal, rather than the offset
index
            permutation quietly producing geometrically meaningless
            comparisons.
            Structural: no pathing, so this lands with rows 1-2 (§4.11).
  (T-SCN-10 reserved: verification of a HORIZONTAL mirror declaration on
an odd
   row count. In axial, after the T-SCN-05 conversion, that map is
       mu(q, r) = (q + r - (H-1)/2,  H-1-r)
   with the ODD row count as the precondition — (H-1)/2 is an integer
exactly
   when H is odd, mirroring rot180's even-H precondition, and the two are
   therefore mutually exclusive on one map. It is a true isometry, not an
index
```

```
   permutation: it sends (dq, dr) to (dq+dr, -dr), which permutes the
three
   terms of the axial metric and leaves the distance unchanged. Blocked on
Q26;
   no gate is written until the schema admits the value.)
Determinism: pure parse + validation; any failure refuses the whole file
with a
         reason. scenarioHash is platform-stable by canonical ordering.
The
         T-SCN-06/08 lane costs are Stub-3 path costs and inherit its
         determinism (T-MOVE-04's canonical tie-break, T-MOVE-06), so the
         reported integers reproduce across runs and compilers.
Acceptance: T-SCN-01..09 headless. The §4.2 validate_scenario MCP tool
wraps the
         same checks in-editor for the Content agent; its manual fallback
is
         running the headless validator on the exported file (MCP stays
off the
         critical path, §3 guardrails).

SPEC STUB 8: UI binding contract           (Director → UI Scaffolder)
Scope:   NOT layout or visual design (§2.11's lane) — this is
         the contract for how every widget is fed. Widgets bind to a view-
model
         snapshot plus the §4.9 event list, and hold no rules state
(§4.1).
Snapshot fields (read-only, produced by the rules module):
         per-hex   {terrainId, owner}
         per-unit  {id, side, unitId, hex, hp, hpMax, isFlag, hasActed,
                    captureProgress}
         per-side  {fameTotal, fameCombat, objectivesHeld X of N,
survivingHP}
         match     {turn, turnCap, sideToMove, resultTier or null}
Queries: reachable(unit) → the T-MOVE-01 set; forecast(attacker,
defenderHex) →
         {damage, counterDamage} computed by the verified resolveDamage /
         defenderCanCounter (5ffa8d6).
Invariants:
  T-UI-01  forecast = resolution: the forecast shown before commit is
produced
           by the same strat call that resolves the attack — identical
numbers,
           mechanically (§2.6, §2.11)
  T-UI-02  the reachable-hex highlight displays exactly the T-MOVE-01 set
— the
           UI queries the module and never recomputes movement (§2.5)
  T-UI-03  the live standings scoreboard (§2.11, §2.8) binds 1:1 to
snapshot
           fields — enemy strength destroyed, objectives held X/N,
surviving
           units/HP, turn vs cap — with no widget-side arithmetic
  T-UI-04  the production menu binds to the buildlist derived from the
four
           Stub-2 unit rows plus current fameTotal; the flag never appears
           (T-SCN-01's non-producible clause, enforced at the UI layer
too)
Determinism: widgets are pure functions of snapshot + events; asserted
         end-to-end by T-INT-05 (§4.9).
Acceptance: T-UI-01..02 headless (the queries are headless functions);
         T-UI-03..04 in-editor Automation.
```

Open questions (Director rulings owed). Every gap found while writing the §4.7 gates (Q1–Q10), the stage-2 additions (Q11–Q13), the rules- and scenario-side rulings folded in here (Q14–Q20), the two gaps found while gating §2.13.1's opening-capture invariant (Q21–Q22), the milestone contradiction the document knowingly carries (Q23), the two raised by the §2.13 symmetry correction (Q24–Q25), and the one raised by correcting that correction (Q26), and the guided opening's one input-gating constraint (Q27) — so that each question carries exactly one ID across the whole document. Each blocks the gate named beside it; the Director writes the rule, the gate then pins it. The last column is not uniform, and the difference matters: **where a reading is stated, it is the conservative one, and it is what ships and what the gates assert** — chosen so that a later ruling loosens behavior rather than invalidating a passing gate. **Rows marked *unruled* state no reading and block their gate outright** (Q4's interruption semantics, Q5's stacking, Q6, Q8, Q9's target- and build-choice ties, and Q20); those gates and milestones cannot be settled until the Director answers.

ID	Question	Blocks	Assumption in force until ruled
[illegible]	Map dimensions. §2.2 and §2.10 never state the	T-HEX-05	Bounds are per-scenario data, not a global constant;

ID	Question	Blocks	Assumption in force until ruled
Q1	prototype map's size or shape.	bounds; Stub 7's `map` field	*Ferrum Crossing* ships 1×9 (§2.13.2).
Q2	Movement classes. §2.3's caption says move cost is "per movement class" and §2.4 gives Recon "ignores some terrain cost," but no class set or per-class cost table exists.	§4.8's `MoveClass` column; the reserved T-MOVE-07	§2.3's single cost column applies to every land unit; no Recon discount is implemented, and no gate is written until the rule exists.
Q3	Pass-through. §2.5 pins one-unit-per-hex for *ending* a move; it is silent on pathing *through* a friendly-occupied hex.	T-MOVE-03	Blocked — a unit may not path through any occupied hex, friendly or not.
Q4	Capture N and interruption. Pin N (§2.7 says "start N=1–2"), and rule the edge cases: does progress reset if the Infantry leaves or dies mid-capture?	T-FAME-05's exactness; the §2.9 AI capture step	N = 1 on the shipped scenario (§2.13.3's recommendation, inside §2.7's stated range). Interruption semantics are **unruled**, so T-FAME-05 stays blocked.
Q5	Kill-award exact values. "~half its Fame cost" (§2.7) is not assertable. Does the flag's +500 stack with the Tank's ordinary kill award?	T-FAME-07	The §2.7 change request proposes exactly half of each §2.4 cost — 50 / 75 / 100 / 150, all integers. Stacking is **unruled**.
Q6	Undamaged-strike bonus. "A small bonus" (§2.7) has no number, yet it feeds the tiebreak's primary key. Options: (a) price it; (b) keep it in the Fame pool but exclude it from the cap tally until priced; (c) cut it — kills already pay half-cost, and the positional triangle already rewards a clean standoff strike with tempo.	T-FAME-07; the T-CAP- tally suite; the kb economy block	**Unruled** — unpriced, so no gate asserts it. The rules author recommends (c), with (b) as the fallback; either unblocks the T-CAP- suite immediately.
Q7	~~Turn cap value.~~ **RULED (this revision).** The cap is **per-scenario data**, stored in Stub 7's `turnCap`; *Ferrum Crossing* ships **20** turns.	—	Ruled. §2.8, §2.11.4 and §2.13.2 now agree, and T-TURN-04 and Stub 7 read a value rather than an example.
Q8	Income timing and build limits. When within the turn does factory/town income pay, and can it fund a build the same turn? Is there a builds-per-factory-per-turn limit (§2.9's AI implies one; the player's rule is unstated)? For a waiting build (§2.7 "the build waits"), is Fame committed at queue time or spawn time, and can it be canceled?	T-FAME-02 and T-FAME-04	**Unruled** — all three sub-questions. T-FAME-02 and T-FAME-04 stay blocked until ruled, and they gate Stub 4, which §4.11 builds *before* the turn loop.
	AI tie-breaks. §2.9's "fl…	T-AI-06; AI determinism as a	Position ties break by canonical hex order (as gated). Target-choice and

ID	Question	Blocks	Assumption in force until ruled
Q9	AI tie-breaks. §2.9's "best expected-damage target" and pathing choices can tie.	stated rule rather than an implementation accident	build-choice tie order are unruled and need naming so determinism is a rule, not an accident.
Q10	Flag designation. Confirm exactly one flag per side, designated by the scenario (`isFlag`, Stub 7), with otherwise-standard Tank stats.	T-SCN-01's exactness; the flag's §4.8 representation	§2.4's "Tank variant" reads as *not producible, and nothing else.*
Q11	Undo. The command log makes single-step undo of an uncommitted move nearly free, but the GDD never grants it, and §2.6's forecast-then-commit flow arguably forbids it.	Nothing today — no gate assumes undo exists, and §2.11.1's `Z` binding is explicitly conditional	No undo: a completed move stands.
Q12	Zero-RNG confirmation. §2.6 and §4.1 say any RNG "is seeded"; nothing in §2 actually uses RNG.	§4.10's `seed` field; T-SAVE-01/02 if RNG is ever added	The prototype ships with **none**, so `seed` is a reserved field written as 0.
Q13	Player-facing replay. §4.10's format makes a "watch replay" feature cheap, but §2.10 does not scope one.	The §2.10 scope table; a UX handoff if it is scoped	The format stays internal — saves, gates, and balance logs only.
Q14	Capture-in-progress at the cap. Does a partially captured objective count toward "objectives held" (§2.8 criterion 2)?	T-CAP-05; the kb victory table	It counts for nobody until the objective flips — §2.7's flip-on-capture wording grants nothing before the flip. Partial credit would need a fractional-count rule and would invert T-CAP-05.
Q15	The 5-unit standard starting force (§2.13.1) — 1 Flag Tank, 2 Infantry, 1 Artillery, 1 Recon — has no antecedent outside §2.13, alongside map dimensions and town counts.	§2.13.1; Stub 7's deployment list	As drafted in §2.13.1: one of each producible system live from turn 1, a 550-Fame producible force.
Q16	Recon/Air vs. Water. §2.3 marks Water passable by "sea, air"; §2.4's "Recon/Air" never says whether it *is* air. If Recon crosses Water freely, every bridge chokepoint and *The Causeway*'s lockout premise leaks.	All three §2.13 maps; the terrain schema	Recon is a **land** unit with terrain-cost discounts, and bridges bind it. All three maps are priced on this reading.
Q17	Cross-Water Artillery fire. With LOS blocking a stretch goal (§2.2), bank-to-bank fire across a one-hex river is legal at ship, and both river maps price a bridge lock on it.	§2.13.2 and §2.13.6 balance, if LOS ever ships	Legal. If LOS blocking ships, Water must not block — or those two maps need a redesign pass.
Q18	Seat-select scope. Is choosing your seat on the shipped map in scope as "scenario data + one menu affordance"?	§2.13.4's replay ceiling — 6 configurations if yes, 3 if no	In scope, as §2.13.4 assumes. It is the cheapest replay lever in the document.
	Factory count as a per-scenario dial. *Longwater March* (§2.13.5) ships **6** factories against §2.7's		Yes — ~4 describes *Ferrum*

ID	Question	Blocks	Assumption in force until ruled
Q19	factories against §2.7's "typical ~4 factories total" — inside §2.7's "two or more neutral" clause, but above its stated typical. Does §2.7's ~4 describe the shipped map only, leaving count as the match-length dial §2.13.5 uses it as?	§2.13.5; T-SCN-03's economy check, which currently cites "~4 factories total"	*Crossing*: §2.13.5's 6 is a deliberate long-map dial, and T-SCN-03 asserts only the home-plus-two-neutral floor.
Q20	Save/replay milestone split. §4.11 shows §4.4's week-5 save/load placement is one week late in one respect: the format and headless replayer are the instrument for the week-2 integration gate (T-INT-02) and the week-4 self-play logs (T-SAVE-07). Split the row — format + replayer early, save-slot UI stays week 5?	§4.4's milestone table; T-INT-02 and T-SAVE-07 sequencing	Unruled. §4.4 stands as written; §4.11 records the conflict without resolving it. This is a scheduling decision, adjacent to **Q23**.
Q21	Opening-capture lane measurement. Does T-SCN-06 price the lane on **terrain alone**, or on the board **as deployed** — where, under Q3's blocked-pass-through reading, a seat's own four other starting units can make its own lane unmeasurable? The two readings can disagree by several MP on a crowded deployment.	T-SCN-06's pass/fail and T-SCN-08's reported integers; §2.13.1's three-map lane table if the answer is "as deployed"	Terrain alone, occupancy excluded — the reading that reproduces §2.13.1's measured 5/5, 4/4, 3/3, and the reading that matches how the lane is actually played (the other four units move too). **Scope narrowed by the §2.13 symmetry correction:** both stretch lanes are now clear of their own seat's starting units and price identically under either reading, so an "as deployed" ruling could only move *Ferrum Crossing*'s numbers — this is effectively a one-map question.
Q22	Uncontested vs. merely reachable. §2.13.1 promises the guided lane is "uncontested, not merely reachable," but states it as a property of the shipped map rather than a checkable rule. Is *distinct objectives per seat* the whole requirement, or must the validator also assert non-contention — e.g. the opposing seat's cheapest Infantry lane to the same objective is strictly longer?	Whether T-SCN-07's distinctness clause is the floor or a further T-SCN invariant is owed	Distinctness only, as gated in T-SCN-07. Consequence stated plainly: a map can pass every §2.13 gate and still hand both seats a race to the same tile, turning the first lesson into a contest.
Q23	~~Milestone-vs-build-order contradiction.~~ **RULED (this revision).** §4.11's critical path makes the baseline AI (row 6) depend on row 5, which depends on row 4 Capture & Fame — but §4.4 promises a working vertical slice *with* the baseline AI in	§4.4's week 1–2 milestones; §2.10's IN row; §4.11's critical path; and Q20, which is the same	Ruled: the vertical-slice milestone moves to **week 3**, and week 2 delivers **move + attack only** (§4.4). §1, §2.10, §4.4 and §4.11

ID	Question	Blocks	Assumption in force until ruled
	slice *with* the baseline AI in week 2, and §2.10 states capture + Fame production “land wk 3, not wk 1–2.” The two schedules cannot both hold.	which is the same decision for save/replay	now describe one schedule — §1's “core playable” line is the fourth site and was corrected with them.
Q24	Symmetry as a declarable value. §2.13.1 previously offered `mirror / rotation / none`, but an odd-r offset rectangle has **no vertical** mirror axis at any dimension, and admits a 180° rotation only when the row count is even. So a vertical `mirror` is a value the validator could never legally accept, and `rot180` is only well-formed against an even-H map. It does, however, have a **horizontal** mirror axis (c ↔ c, r ↔ H−1−r) exactly when the row count is **odd** — the geometry the shipped 11 × 9 map sits on (§2.13.1 fact 1, §2.13.2) — so the field is a choice, not a forced hand. Narrow the field to `rot180 | none` with an even-row-count precondition, and make `rot180` on an odd row count a hard refusal rather than a failed hex comparison? (**Whether a third value should be admitted for the horizontal mirror is Q26**, which owns that question — this row asks only about the narrowing.)	§2.13.1's validation-invariant list; Stub 7's `symmetry` field and T-SCN-08's fixtures (§4.7, `tech-director-owned`)	Narrowed, as §2.13.1 now reads: `rot180 | none`, and `rot180` declared on an odd row count refuses the file with a reason (a wrong dimension is an authoring error, not a balance question). Both stretch maps are redrawn on 8 rows and declare `rot180` (§2.13.5, §2.13.6); *Ferrum Crossing* declares `none`. If the Director rules otherwise, the only consequence is that a bad declaration surfaces as N failed hex comparisons instead of one refusal — no layout moves. Whether the enum should also carry a horizontal-mirror value is answered at **Q26**, not here.
Q25	What a `rot180` declaration binds. §2.13.1 says declared symmetry is machine-verified but never says over *what*. Terrain only? Terrain + ownership + placements — the reading both stretch maps are actually drawn to (§2.13.5 “every one of those is a ρ-pair”; §2.13.6 “East is the exact ρ-image of West”)? And does it bind `guidedOpening`, so the two seats' lanes must themselves be ρ-images?	T-SCN-09's assertion set; and T-SCN-08's fixture (b), whose equal 4 / 4 is a *theorem* if `guidedOpening` is bound and only a *measurement* if it is not	Terrain + ownership (sides exchanged) + placements (sides exchanged), as gated in T-SCN-09. `guidedOpening` is **not** bound: §2.13.1 requires each seat's lane to be its own neutral within 6 MP, never that the two be images of each other, so a `rot180` map may legitimately name non-image lanes and report a split. Costless either way today — both stretch maps satisfy the *stronger* reading as drawn (*Longwater*: ρ(1,2) = (11,5) and ρ(4,1) = (8,6); *Causeway*: ρ(1,2) = (7,5) and ρ(3,2) = (5,5)), so ruling `guidedOpening` in would fail no shipped map, and

ID	Question	Blocks	Assumption in force until ruled
			shipped map, and ruling terrain-only would merely loosen T-SCN-09.
Q26	Is a **horizontal** mirror declarable? Q24 narrowed the enum to `rot180 | none` partly on the ground that `mirror` is a value the validator could never legally accept. That holds for the **vertical** axis at every dimension, but not for the horizontal one: $\mu(c, r) = (c, H-1-r)$ is a genuine isometry of an odd-r rectangle whenever the **row count is odd** — $H-1$ is then even, so r and $H-1-r$ share a parity, the offset is preserved and the column is unchanged. *Ferrum Crossing* is 11 × **9** and carries that axis geometrically (§2.13.1). So Q24's enum survives, but as a **scope** decision, not an impossibility. Add `mirrorH` with an odd-row-count precondition — the exact counterpart of `rot180`'s even-row-count one — or keep the enum at two values and accept that a true horizontal mirror is undeclarable?	Stub 7's `symmetry` field (§4.7); the reserved **T-SCN-10**; nothing else — no gate asserts anything about a value the schema does not admit, and no §2.13 map is drawn to a horizontal mirror	The enum stays `rot180 | none` and T-SCN-10 stays unwritten: adding a value to a REQUIRED enum moves every scenario's hash, and no shipped map needs it — *Ferrum Crossing* declares `none` because its terrain is asymmetric, and both stretch maps are even-H `rot180`. The conservative reading is the narrow one, so a later ruling only ever *widens* the accepted set and *adds* an assertion; nothing that passes today would start failing. The cost of leaving it narrow, stated plainly: an author who genuinely draws a horizontal mirror on an odd-H map must declare `none`, and `none` asserts nothing — a verifiable property is silently discarded rather than an authoring error caught, which is the opposite of the failure mode Q24 was choosing between. Note the two are never in competition: `rot180` needs even H, a horizontal mirror needs odd H, and their composition would be the vertical mirror that never exists, so at most one non-`none` value is well-formed on any given map.
Q27	Is gating End Turn during beat 1a **presentation or rule?** During beat 1a of a guided opening only, End Turn is inert until the marked Infantry (`guidedOpening.infantry`) has moved; hover reads `Move the marked`	Nothing today, and nothing in §4.7: no stub or `T-` ID gates the directive strip, and Stub 7 deliberately keeps the guidance layer out of Stub 8's snapshot. The dependency is	Ships as specified (conservative). If ruled the other way: a player who ends turn 1 without moving leaves 1a outstanding, and rule 1 hands the strip to 1b (`End`

ID	Question	Blocks	Assumption in force until ruled
Q27	`Infantry first`. It is scoped to the first match, dies with `Skip guidance`, and never applies outside the guided window — but it is the only guided-opening constraint that gates a player *input* rather than a selection, which is why it is registered rather than assumed.	internal to §2 — §2.11.6-B's turn-1 row is unconditional in all three branches only if 1a cannot outlive turn 1, so a ruling of "no input gating" would require that row to be re-derived	`turn`) an instruction they have just followed — so the fallback is to let 1a expire silently at the turn boundary like 1b, and §2.11.6-B's turn-1 row gains a footnote.

4.8 Data contract — DataTable schemas

Principle: authored once, read twice, proven equal. Each table is one canonical CSV in the repo (`data/`). The headless loader parses it directly; the Unreal editor imports the same file into a `UDataTable` whose row struct derives `FTableRowBase`. A parity gate (T-DATA-05, Unreal Automation) iterates every imported row and asserts it equals the CSV field-for-field. Nothing is authored twice, so the headless sim and the engine can never disagree about a stat. Missing column or unparseable value = hard load failure, never a silent default.

Unit schema — `data/units.csv` → headless `strat::UnitDef` → UStruct `FUnitRow`. Exactly four rows (Infantry, Tank, Artillery, Recon; §2.4). The **flag unit is not a row**: §2.4 defines it as "a designated Tank," so flag status is a placement-level field in the scenario file (`isFlag`, §4.7 Stub 7), gated by T-SCN-01 — not a fifth unit type. One representation, one gate.

Column	CSV type	Headless field (`strat::UnitDef`)	UStruct field (`FUnitRow`)	Source
`Id` (row name)	string	`id`	RowName (`FName`)	§2.4
`HP`	int	`hpMax`	`int32 HP`	§2.4 (10/20/8/12)
`Move`	int	`move`	`int32 Move`	§2.4 (3/5/3/7)
`Atk`	int	`atk`	`int32 Atk`	§2.4 (4/8/10/5)
`Def`	int	`def`	`int32 Def`	§2.4 (2/5/1/3)
`RangeMin`	int	`rangeMin`	`int32 RangeMin`	§2.4 (Artillery 2, others 1)
`RangeMax`	int	`rangeMax`	`int32 RangeMax`	§2.4 (Artillery 3, others 1)
`CostFame`	int	`costFame`	`int32 CostFame`	§2.4 (100/300/200/150)
`Type`	enum string	`strat::UnitType type`	`EUnitType Type`	addendum Part A — order fixed: Infantry, Tank, Artillery, Recon
`CanCapture`	bool	`canCapture`	`bool bCanCapture`	§2.7 (Infantry only)
`MoveClass`	enum string	*reserved*	*reserved*	**blocked on Q2**

`EUnitType` is a `UENUM` mirroring `strat::UnitType` (`Combat.h`, addendum Part A) with the enumerator order pinned; T-DATA-05 asserts the mirror is exact so an editor-side reorder can never silently reindex the effectiveness table below.

Terrain schema — `data/terrain.csv` → headless `strat::TerrainDef` → UStruct `FTerrainRow`. Exactly seven rows (§2.3).

Column	CSV type	Headless field	UStruct field
`Id` (row name)	string	`id`	RowName
`MoveCost`	int (0 = impassable)	`moveCost`	`int32 MoveCost`
`DefensePct`	int, signed	`defensePct`	`int32 DefensePct`
`PassLand/PassAir/PassSea`	bool ×3	`passLand/Air/Sea`	`bool bPassLand/Air/Sea`
`Capturable`	bool	`capturable`	`bool bCapturable`
`IncomeFame`	int	`incomeFame`	`int32 IncomeFame`
`IsSpawnPoint`	bool	`isSpawnPoint`	`bool bIsSpawnPoint`
`IsRepairPoint`	bool	`isRepairPoint`	`bool bIsRepairPoint`

Type-effectiveness schema — `data/effectiveness.csv` → `strat::effectiveness` → UStruct `FEffectivenessRow`. A 4×4 matrix, row = attacker type, columns = defender types, values ∈ {0.5, 1.0, 1.5} (§3 spec), **shipping all-1.0** (§2.4 — the triangle stays positional). The verified implementation (`Combat.cpp::effectiveness` @ `5ffa8d6`) hardcodes the neutral stub; this schema is the lever's *data* form, so that if self-play ever asks for a non-1.0 cell, populating it is a CSV edit gated by the existing directional-gate plan (addendum Part A), not a code change. T-COMBAT-09 (neutral stub, 16/16 pairs) continues to pin the shipped state; a non-neutral CSV with T-COMBAT-09 still in the suite is a deliberate, visible gate change the Director must approve — the "do not invent balance values" rule enforced by the pipeline itself.

```
Invariants (the T-DATA set — Stub 2, §4.7):
  T-DATA-01  loaded unit values equal the §2.4 table exactly (4 rows, all
columns)
  T-DATA-02  loaded terrain values equal the §2.3 table exactly (7 rows),
             including Bridge's NEGATIVE defense and Water impassable-to-
land
  T-DATA-03  exactly one unit row has CanCapture == true (Infantry, §2.7)
  T-DATA-04  sanity: all costs > 0; RangeMin <= RangeMax; HP > 0
  T-DATA-05  (editor, Unreal Automation) every imported DataTable row
equals the
             CSV field-for-field, and EUnitType mirrors strat::UnitType
exactly
  T-DATA-06  effectiveness.csv is 4×4, indexed in the pinned type order,
every
             cell ∈ {0.5, 1.0, 1.5}; the SHIPPED file is all-1.0 (re-
asserting
             T-COMBAT-09 at the data layer)
Determinism: pure parse; missing/malformed field = hard fail, no defaults.
Acceptance: T-DATA-01..04, 06 headless; T-DATA-05 in the editor pass.
```

4.9 Headless-module → Unreal integration path

The rules module's value is that it has **zero engine dependencies** (§3, §4.1); integration must add Unreal *around* it without ever adding Unreal *to* it.

1. Module layout — one source, two compilers. The certified headless sources (`Combat.h`/`Combat.cpp` today; each §4.7 stub as it lands) live canonically in the crew repo, where the `g++`/`clang++` + `python run.py` gate runs (§3 ledger). The UE project vendors them verbatim into a UBT runtime module, `Source/StratRules/`, via a sync script that records the source commit hash. `StratRules` contains **no engine headers, no UObject, no third-party includes** — pure C++17 in `namespace strat`, exactly the base-spec constraint. The standalone gate keeps compiling the identical files, so "the engine build works" never substitutes for "the gate passed."

2. Bridge — the only code that knows both worlds. The game module (`Stratocracy`) owns: - **Load:** `FUnitRow`/`FTerrainRow`/`FEffectivenessRow` → `strat::UnitDef` / `strat::TerrainDef` / effectiveness table (a mechanical §4.8 mapping), plus `strat::loadScenario` on the shipped scenario asset (§4.7 Stub 7). - **The authoritative `strat::GameState`.** Actors and UMG hold no rules state (§4.1 "never own rules" — here made structural, not aspirational). - **Command in / events out.** Presentation submits commands — `Move{unit, destHex}`, `Attack{unit, targetHex}`, `Build{factoryHex, unitId}`, `Capture{unit}`, `EndTurn{}` — the rules module validates then applies each, and emits an **ordered, deterministic event list**: `Moved(path)`, `Damaged`, `Destroyed`, `Captured`, `Spawned`, `BuildWaiting`, `Repaired`, `IncomePaid`, `MatchEnded(tier)`. Actors and widgets animate and rebind **from events and the view-model snapshot only** (§4.7 Stub 8). An invalid command returns a rejection reason and changes nothing. - **Threading:** synchronous on the game thread. A full turn resolution is microseconds of integer math; the shipping AI "moves instantly" (§2.8). No async, and no MCP involvement anywhere in the runtime path — the MCP plugin remains editor-only tooling, experimental, off the critical path (§3 guardrails, §4.2, §4.5).

3. Parity gates. The command log format of §4.10 is the instrument: the same recorded match is replayed by the headless harness and by an in-editor Automation test through the UBT-compiled module, and both must land on the same canonical state hash. This is what catches the one genuinely engine-shaped risk — a compiler/CRT divergence in the damage formula's `round` — mechanically instead of by playtest anecdote.

```
SPEC STUB: Integration parity              (Director → Systems Engineer /
UI Scaffolder)
Inputs:  vendored StratRules sources + recorded source commit; a §4.10
replay
         file; the §4.8 tables imported in-editor.
Invariants:
  T-INT-01  source identity: every file in Source/StratRules/ hash-matches
the
            recorded crew commit — the ledger's evidence chain survives
vendoring
  T-INT-02  replay parity: the same command log replayed headless and in-
engine
            (Automation test) produces the same final canonical state
hash.
            (The tripwire of this stub: an agent that "ports" rather than
vendors
            the module — or a compiler that rounds differently — passes
every
            behavior test in one world and silently diverges in the
other.)
  T-INT-03  rejection safety: an illegal command leaves the state hash
unchanged
            and returns a reason; no partial application
  T-INT-04  no engine deps: StratRules compiles standalone under the
existing
            g++/clang++ gate — the gate run itself is the assert
  T-INT-05  (editor, Automation) presentation statelessness: after any
event
            sequence, rebuilding all widgets/actors from the current view-
model
            snapshot alone reproduces the same displayed values (nothing
lives
            only in a widget)
Determinism: the bridge never reorders, drops, or synthesizes events.
Acceptance: T-INT-01, 04 on every gate run; T-INT-02, 03, 05 in the editor
pass.
```

4.10 Save & replay format

Design choice: a save *is* a replay. Because every system is a deterministic pure function or state machine (§2.6, §4.1, and every §4.7 determinism gate), the cheapest correct save is not a state snapshot but **the scenario reference plus the ordered command log**. Loading = `loadScenario` + re-apply the log — headless-speed, sub-second. One format, four consumers:

1. **Single-slot save/load** (§4.1's "minimal single-slot," now specified).
2. **The T-INT-02 parity gate** — its input file is a save file.
3. **Balance Analyst self-play logs** (§4.1 harness) — each self-play match is emitted as

this same format, so every balance claim in the ledger's evidence chain is a *replayable* artifact, not a summary statistic.
4. **Bug reproduction** — a playtest failure attaches its save; any agent can replay it headless.

File layout — versioned JSON, one file:

Field	Type	Meaning
`formatVersion`	int	This layout's version; unknown = refuse load
`rulesCommit`	string	Crew commit of the rules module that wrote the file (T-INT-01's hash)
`dataHash`	string	Hash of the §4.8 CSV set in effect
`scenarioId` / `scenarioHash`	string	The §4.7 Stub-7 scenario file and its hash
`seed`	int	Reserved; **written as 0** — no RNG ships (§2.6, pending Q12)
`commandLog`	array	The ordered commands of §4.9, exactly as the bridge consumed them, tagged `{turn, side}`
`stateHash`	string	Canonical hash of the resulting state (integrity check)
`result`	string/null	Result tier (§2.8) if the match ended, else null

Canonical state hash. Defined once, in the headless module: serialize the `GameState` in a fixed field order — turn counter, side to move, per-side `fameTotal`/`fameCombat`, objective ownership, per-unit `{id, side, hex, hp, isFlag, captureProgress, pendingBuilds}` sorted by the canonical hex order (§4.7 conventions, T-HEX-07) — then hash the bytes. Every field is an **integer** (`eff` and the HP ratio exist only transiently inside `resolveDamage`), so the hash is platform-stable by construction; T-INT-02 proves it across compilers.

Policies (prototype): - **Single slot** — one file (`slot0`), stored via a thin `USaveGame` wrapper holding the JSON so platform save paths are respected. Overwrite-confirm UX is **unowned**: §2.11 specifies no save/load surface, and neither its screen list (§2.11.5) nor its build ranking (§2.11.8) includes one — either §2.11 gains the surface, or the prototype ships silent single-slot overwrite. - **Save points** — a save is accepted between atomic commands during the player's turn. The AI turn resolves synchronously in one call (§4.9), so "mid-AI-turn" is not a reachable save state by construction. - **Mid-match saves** — the log up to the last completed command *is* the save. Derived pending state (a waiting build, capture-in-progress) replays correctly because it is a function of the log. - **Version policy** — mismatched `formatVersion`, `rulesCommit`, `dataHash`, or `scenarioHash` → **refuse load with a reason**. No migration in the prototype; a save is only valid against the exact rules and data that wrote it.

```
SPEC STUB: Save & replay                   (Director → Systems Engineer)
Inputs:  scenario file (Stub 7), command log, §4.8 data set, module
commit.
Invariants:
  T-SAVE-01  round-trip: play N commands → save → load → identical
stateHash
  T-SAVE-02  replay determinism: the same file loaded twice → identical
hashes
             (leans on every §4.7 determinism gate — T-HEX-07, T-MOVE-06,
             T-FAME-09, T-TURN-09, T-AI-06; this is their end-to-end
composition)
  T-SAVE-03  prefix validity: every prefix of a valid log is itself a
valid,
             loadable save (mid-match save falls out of this)
  T-SAVE-04  refusal: any header mismatch (version/rules/data/scenario
hash) →
             load refused with a reason; state untouched
  T-SAVE-05  no partial load: a log with an illegal command at index k is
             refused whole; the pre-load state survives. (The tripwire: an
agent
             that applies-then-validates leaves a corrupted half-loaded
state
             that passes every happy-path test.)
  T-SAVE-06  stateHash stability: hash of a given state is identical
across the
             headless and in-engine builds (asserted jointly with T-INT-
02)
  T-SAVE-07  harness compatibility: a Balance Analyst self-play log
validates
             and replays as a save file — one format, no dialect drift
Determinism: pure; the file contains everything needed to reproduce the
state.
Acceptance: T-SAVE-01..07 headless (05 also exercised in-editor via the
load UI
            path); slot I/O smoke test in the editor pass.
```

4.11 Build order

Rows 1–8 are the §4.7 stubs (the eight `*pending*` ledger rows); rows 9–10 are the §4.9 and §4.10 systems. Combat, its test suite, Repair, and Type-effectiveness are green at `5ffa8d6` and are prerequisites, not work items. Note **row 10's format spec must exist by the row-9 integration pass**, because T-INT-02's input file is a §4.10 save.

#	System (ledger row)	Depends on	Headless?	Acceptance test IDs
1	Hex grid & math (§4.7 Stub 1)	— (Q1 pins bounds)	Yes	T-HEX-01..07
2	Data tables (§4.8, incl. effectiveness CSV)	— (MoveClass column blocked on Q2)	Loader + T-DATA-06 yes; import parity in-editor	T-DATA-01..06
3	Movement & pathfinding (Stub 3)	1, 2	Yes	T-MOVE-01..06
4	Capture & Fame economy (Stub 4)	3	Yes	T-FAME-01..09
5	Turn loop & win/tiebreak (Stub 5)	4 + verified Combat/Repair @ 5ffa8d6	Yes	T-TURN-01..09
6	Opponent AI (Stub 6)	5	Yes	T-AI-01..06 + self-play smoke
7	Scenario file & validator (Stub 7)	1, 2 for the structural half (T-SCN-01..03, 05, 07, 09); **3 for the priced half** — T-SCN-04, 06, 08 all cost a path	Yes; MCP tool wraps it in-editor, manual fallback stands	T-SCN-01..09
8	UI binding (Stub 8)	5, 7 (snapshot needs full state)	Contract + queries yes; widgets in-editor	T-UI-01..04
9	Presentation bridge & integration — §4.9 (proposed ledger row)	Rows 1–5 built; vendoring + T-INT-01/04 can start with any subset	Source/compile gates yes; replay parity + statelessness in-editor	T-INT-01..05
10	Save & replay — §4.10 (proposed ledger row)	4, 5 (command set + turn loop); 7 (scenario format); **format spec itself has no deps — write it first**	Yes, all but slot I/O	T-SAVE-01..07

Critical path: 1 → 3 → 4 → 5 → 6/8. Row 2 runs in parallel immediately. **Row 7 no longer sits beside the chain; it straddles it.** Its structural half (T-SCN-01..03, 05, 07, 09) starts once 1–2 land, but three of its nine invariants — T-SCN-04's flag reachability and T-SCN-06/08's opening-capture lane — price a Stub-3 path, so row 7 cannot *close* until row 3 does. Row 7 is still not ON the critical path (nothing in the chain waits on it), but scheduling it as "parallel from week 1" would leave its ledger row un-flippable and the §2.11.6 guided opening ungated for however long movement slips: the scenario row flips after movement, not before. 6 and 8 fork after 5. §4.4's milestone table currently parks save/load in week 5, and §4.10 shows that is one week too late in one respect: the *format and headless replayer* are the instrument for the week-2 integration gate (T-INT-02) and the week-4 self-play logs (T-SAVE-07). Splitting the row — format + replayer early, save-slot UI stays week 5 — is filed as **Q20** (§4.7) against §4.4's milestone table, not applied here.